SAFI INSTITUTE OF ADVANCED STUDY (AUTONOMOUS)

Affiliated to University of Calicut, Accredited by NAAC with A++ Grade(3.54)

Recognized by UGC under 2(f),ISO 9001-2015 Certified by AAB Vazhayur East, Malappuram Dist, Kerala– 673633

Department of Computer Applications

Project Report

COLLEGE VOTTING SYSTEM

Done by,

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2. **FIDA FASAL P - (SIAWBCA017)**
3. **AYSHA LABEEBA PP - (SIAWBCA036)**
4. **ANAMIKA CK - (SIAWBCA034)**

Under the Guidance of

Mrs . Safeena C

2022-2025

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SAFI

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CERTIFICATE

Certified that this is the bonafide report on the project work entitled

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During the academic year 2022-2025 in partial fulfilment of requirement for the award of the degree of,

Bachelor of Computer Applications

Of

University of Calicut, Malappuram Dist., Kerala

Internal Guide **Head of the Department**

Internal Examiner **External Examiner**

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Date: 18 March 2025

Place: Vazhayur

TO WHOM IT MAY CONCERN

This is to certify that **Mrs. FATHIMA FIDA KT (REG.NO: SIAWBCA016)**, **Mrs. FIDA FASAL P (REG.NO: SIAWBCA017), Mrs. ANAMIKA CK (REG.NO: SIAWBCA034), Mrs. AYSHA LABEEBA PP (REG.NO: SIAWBCA036)** a 6th semester student pursuing Bachelor of Computer Application (BCA) at SAFI Institute of Advanced Study (Autonomous), Vazhayur, Malappuram has successfully completed their academic project on **COLLAGE VOTTING SYSTEM** using **PHP** from December 2024 to March 2025. Throughout the duration of their academic project, they demonstrated punctuality, diligence, and curiosity. We wish them every success in their future endeavors.

Principal

ACKNOWLEDGEMENT

First and foremost, we express our deepest gratitude to the Almighty God for His abundant blessings, guidance, and grace that have illuminated our path and granted us strength and resilience throughout our academic pursuits.

We are immensely grateful to the Principal, Prof. E. P. Imbichi Koya for their visionary leadership, unwavering support, and commitment to fostering an environment of excellence and innovation within our institute.

Our sincere thanks go to the Head of the Department, Mr. Abdul Samad C. for their invaluable guidance and encouragement, which have been instrumental in nurturing our intellectual curiosity and facilitating our academic growth.

We extend our heartfelt appreciation to our Project Guide Mrs. Safeena C for his/her wisdom and guidance have been invaluable assets in our academic journey.

We would also like to express our gratitude to our teachers for their commitment has been instrumental in shaping our understanding and appreciation of the field of Computer Science. To our families, whose unwavering love, encouragement, and sacrifices have been the bedrock of our support system, we offer our deepest appreciation and gratitude.

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Fathima Fida KT

Fida Fasal P

Aysha Labeeba PP

Anamika CK

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1 ABSTRACT

COLLEGE VOTING SYSTEM

A College Voting System is a digital platform designed to streamline the voting process in educational institutions. This system allows students to cast their votes securely and efficiently through an online platform also reducing the complexities of manual voting.

In the existing system, voting is conducted manually using paper ballots, which is time-consuming and prone to human errors. Counting votes manually often leads to inaccuracies and delays in declaring results. Additionally, there is a lack of transparency and security, This can lead to disagreements and undermine trust in the voting process.

The proposed system addresses these issues by introducing an online voting platform using HTML, CSS, JAVA SCRIPT, and BOOTSTRAP for the frontend, and PHP and MySQL for the backend. This system enables students to log in securely, cast their votes with ease, and view real-time election results. The system ensures secure data handling, faster counting, and improved accuracy.

This online voting system provides several advantages, including a user-friendly interface, faster and more accurate vote counting, and a secure and transparent process. By leveraging modern web technologies, the system aims to make the voting process more reliable and efficient.

Technologies Used:
Frontend – HTML, CSS, JAVA SCRIPT, BOOTSTRAP
Backend – PHP
Database – MySQL

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2. INTRODUCTION

INTRODUCTION

2.1 Overview of the system

The **College Voting System** is an advanced digital platform designed to streamline student elections, ensuring a fair, efficient, and transparent voting process. Traditional paper-based elections are often slow, error-prone, and difficult to manage, leading to delays in result announcements and potential miscalculations. This system eliminates those inefficiencies by providing a secure online platform where students can easily log in, select their candidates, and submit their votes within minutes. By automating the entire process, the system ensures accuracy, reduces the possibility of human error, and enhances trust in the election outcomes.

One of the key strengths of the system is its **real-time vote counting and result display**. Unlike manual counting methods that can take hours or even days, this digital platform instantly tallies votes and presents the results as soon as the voting period ends. The system also features **role-based access**, allowing administrators to manage elections while restricting students to only casting their votes. Additionally, it incorporates robust **authentication mechanisms** to prevent duplicate voting and unauthorized access, ensuring that every vote is counted fairly and securely. The integration of technologies such as **PHP, MySQL, JavaScript, and Bootstrap** ensures a user-friendly experience while maintaining high levels of security.

Beyond efficiency, the **College Voting System** significantly increases student participation in elections. Since voting can be done from any device with an internet connection, students no longer need to visit physical polling stations, making the process more convenient and accessible. Moreover, by reducing the reliance on paper ballots, the system promotes environmental sustainability while cutting down administrative costs associated with traditional elections. Overall, this system modernizes the voting experience, making college elections smoother, faster, and more reliable for both students and administrators.

SYSTEM ENVIRONMENT

System environment is the term commonly used to refer to support an application. A software environment for a particular application could include the operating system, the data base system, specific development tool or compiler.

2.2 Software configuration

For the proposed system to work properly, it is necessary that following software are installed and running on the server / client.

Operating system	: Windows 7 or above / Linux New
Versions Language	: HTML,PHP, JAVA SCRIPT
IDE	: Visual Studio code
Front end	: HTML, CSS, JAVA SCRIPT AND BOOTSTRAP
Other tools	: WAMP, Figma
Back end	:PHP,MYSQL

2.2.1 HTML

HTML (Hyper Text Markup Language) is used to create web pages and web applications. It is a markup language. By HTML we can create our own static page. It is used for displaying the data not to transport the data.

2.2.2 PHP

PHP (Hypertext Preprocessor) is a server-side scripting language used for developing dynamic web pages and handling backend processes. It can connect to databases, process form data, and generate dynamic content.

PHP can generate dynamic page content

* PHP can create, open, read, write, delete, and close files on the server

* PHP can collect form data

* PHP can send and receive cookies

* PHP can add, delete, modify data in your database

* PHP can be used to control user-access

* PHP can encrypt data

2.2.3 CSS

Cascading Style Sheets (CSS) is a stylesheet language used to describe the presentation of a document written in HTML or XML (including XML dialects such as SVG, MathML or XHTML). CSS describes how elements should be rendered on screen, on paper, in speech, or on other media.

2.2.4 JAVA SCRIPT

JavaScript is a versatile, high-level programming language used for building dynamic and interactive web applications. It runs on the client-side (browser) but can also be used on the server-side with Node.js. JavaScript enhances web pages by enabling real-time updates, form validation, and event handling. It supports Object-Oriented Programming (OOP) concepts like abstraction, inheritance, and encapsulation. Libraries and frameworks like

React.js, Vue.js, and jQuery simplify development. JavaScript is essential for creating responsive, fast, and interactive web applications, including a college voting system.

2.2.5 BOOTSTAP

This Bootstrap tutorial is designed for beginners and experienced professional, covering basics and advanced concepts including working with Bootstrap's CSS classes, incorporating JavaScript plugins, and more. This tutorial will help you with the knowledge and skills to create stunning and responsive websites

2.2.6 MYSQL

MySQL is an open-source relational database management system (RDBMS) used for storing, organizing, and managing data. It is based on the **Structured Query Language (SQL)**, which allows users to create, modify, and retrieve data from databases efficiently. MySQL is widely used in web development due to its reliability, scalability, and ease of integration with programming languages like **PHP**.

2.3

Hardware configuration

It is recommended that for optimal performance, the following minimum hardware are installed.

Processor : Intel Pentium IV or above

Monitor : Minimum 14-inch

display RAM : 4 Gb

Hard Disk : 80 GB

Keyboard : Standard 104

Keys Mouse : Serial mouse

3. SYSTEM ANALYSIS

SYSTEM ANALYSIS

System analysis is the process of understanding and defining the requirements and structure of the College Voting System. It involves identifying the existing problems in the current voting system, analyzing the needs of the users, and designing a solution that improves the overall voting process. System analysis helps to ensure that the proposed system meets the functional and non-functional requirements while addressing the limitations of the existing system.

3.1 Existing System

In most colleges, voting is conducted using traditional paper-based methods. This process involves distributing paper ballots, collecting them, and manually counting the votes. The existing system has several limitations:

- **Time-consuming:** The manual counting of votes takes a significant amount of time, delaying the announcement of results.
- **Human Error:** Mistakes can occur during vote counting, leading to inaccurate results.
- **Security Issues:** Paper ballots can be lost or manipulated, compromising the integrity of the election process.
- **Transparency Issues:** The lack of a clear and transparent process can lead to mistrust among students and candidates.
- **Limited Accessibility:** Students may face difficulties in participating in the voting process due to physical constraints or mismanagement of voting materials.

3.2 Proposed System

The proposed College Voting System is designed to overcome the limitations of the existing system by introducing an online platform for voting. The system allows students to cast their votes electronically through a secure and user-friendly interface. The proposed system includes the following features:

- **Secure Login:** Students are required to log in using their unique credentials to prevent unauthorized access.
- **Candidate List:** Students can view the list of candidates and their details before casting their votes.

- **Automated Counting:** The system automatically counts votes in real-time, reducing the chances of human error.
- **Instant Results:** Results are calculated and displayed immediately after the voting period ends.
- **Audit Trail:** The system maintains a record of all voting activities to ensure transparency and allow for verification.
- **Accessibility:** The system can be accessed from any device with an internet connection, making it convenient for students.
-

3.3 Feasibility Study

A feasibility study is conducted to evaluate whether the proposed College Voting System is practical and achievable within the available resources and constraints. It helps to determine if the project is worth pursuing by analyzing its technical, operational, and economic aspects.

3.3.1. Technical Feasibility:

Technical feasibility evaluates whether the proposed system can be developed and implemented using available technology and resources.

- The system will be developed using **PHP** for backend development and **MySQL** for database management.

- **HTML, CSS, JAVA SCRIPT and Bootstrap** will be used for creating a responsive and user-friendly frontend interface.
- The system will be hosted on a local server using **WAMP**, which provides an environment to run PHP and MySQL.
- The technologies chosen are well-supported and widely used, making them easy to integrate and maintain.
- The system is designed to work on major browsers like **Chrome**, **Firefox**, and **Edge**, ensuring broad accessibility.

3.3.2. Operational Feasibility:

Operational feasibility examines whether the system will function effectively and meet user requirements.

- The system is user-friendly, with a simple interface that allows students to log in, select candidates, and cast votes easily.
- It reduces the workload for administrators by automating vote counting and result generation.
- The system ensures secure and transparent voting, which builds trust among students and election organizers.
- Minimal training is required for both students and administrators to operate the system effectively.

3.3.3. Economic Feasibility:

Economic feasibility evaluates whether the project is financially viable and cost-effective.

- The development cost is low because the system is built using open-source tools (PHP, MySQL, Bootstrap).
- The need for physical resources (like paper ballots and counting staff) is eliminated, reducing administrative costs.
- Maintenance costs are low since open-source technologies are used, and updates can be made easily.
- Automation of vote counting and result generation reduces labor costs and human error, improving efficiency.

3.3.4. Scheduling Feasibility:

Schedule feasibility evaluates whether the project can be completed within the given time frame.

- The development of the College Voting System is estimated to be completed within **2–3 months**.
- The system will be developed in phases, including design, coding, testing, and deployment.
- The timeline is realistic considering the scope of the project and the availability of development resources.

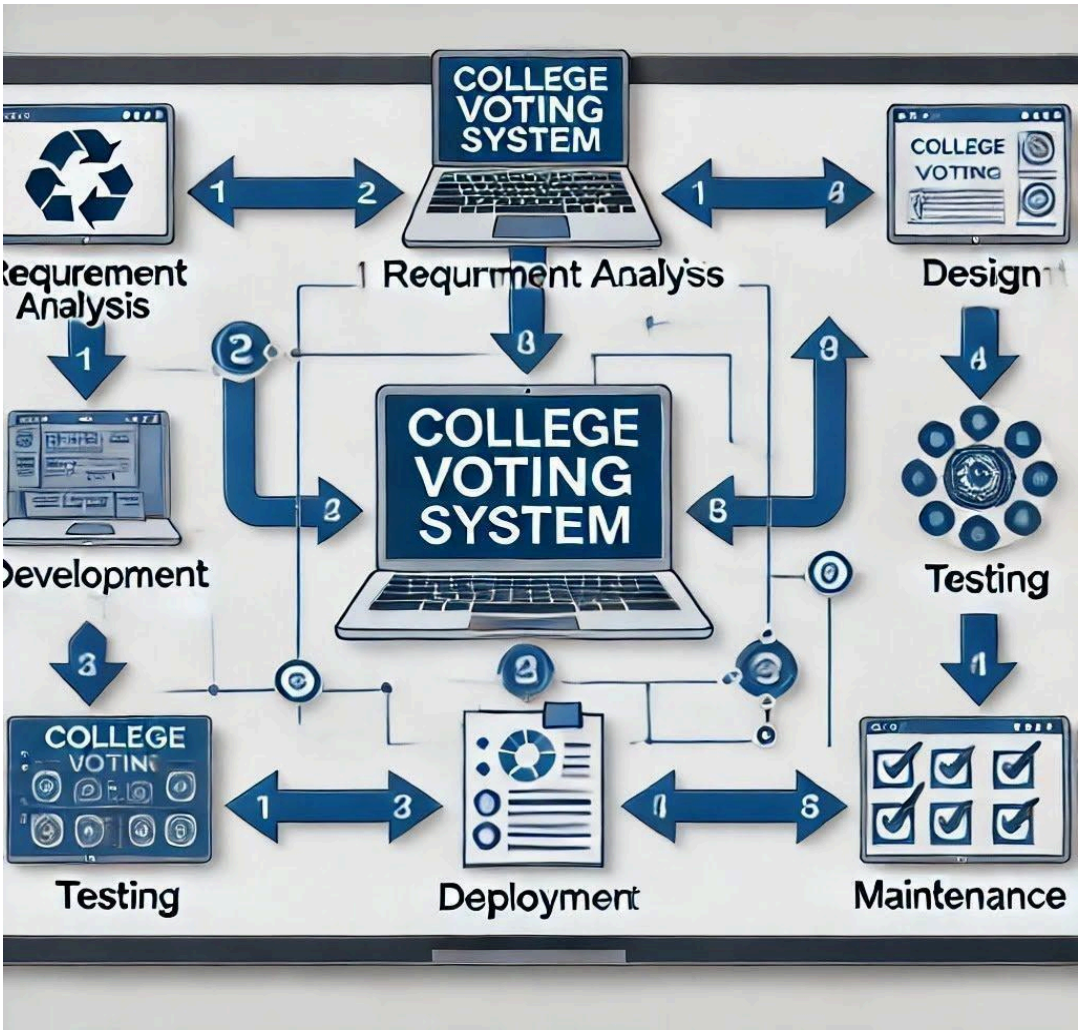
3.4

Project planning

Project planning is a critical phase in the development of the College Voting System. It involves setting clear objectives, defining the project scope, identifying deliverables, allocating resources, and managing potential risks. The primary objective of the College Voting System is to create a secure and user-friendly platform for conducting elections within the college environment. The system will automate the voting process, ensuring accuracy, transparency, and quick result generation. The project scope includes user registration, candidate display, secure voting, and real-time result generation, while external voting and offline functionality are excluded from the scope.

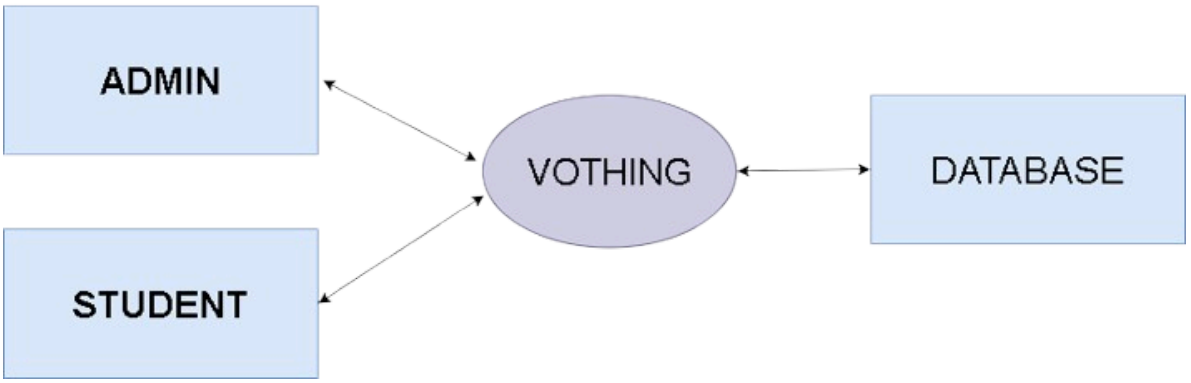
The project is planned to be completed within **12 weeks**, divided into key phases such as **requirement analysis, design, development, testing, deployment,** and **maintenance**. Resources like project managers, frontend and backend developers, and database administrators will be involved in the project. Tools such as **PHP, MySQL, HTML, CSS, JAVA SCRIPT , BOOTSTRAP,** and **WAMP** will be used to develop and manage the system. These technologies ensure that the system will be fast, secure, and scalable, capable of handling multiple users simultaneously without performance issues.

Risk management strategies are essential to the success of the project. Measures include securing user data, preventing unauthorized access, and controlling project scope to avoid delays and cost overruns. The estimated cost for the project is minimal as it relies on open-source tools, which reduces development and maintenance costs. Success will be measured by the system's ability to securely handle votes, provide accurate results, and
ensure user satisfaction. Effective project planning will ensure the College Voting System is delivered on time, within budget, and meets user expectations.

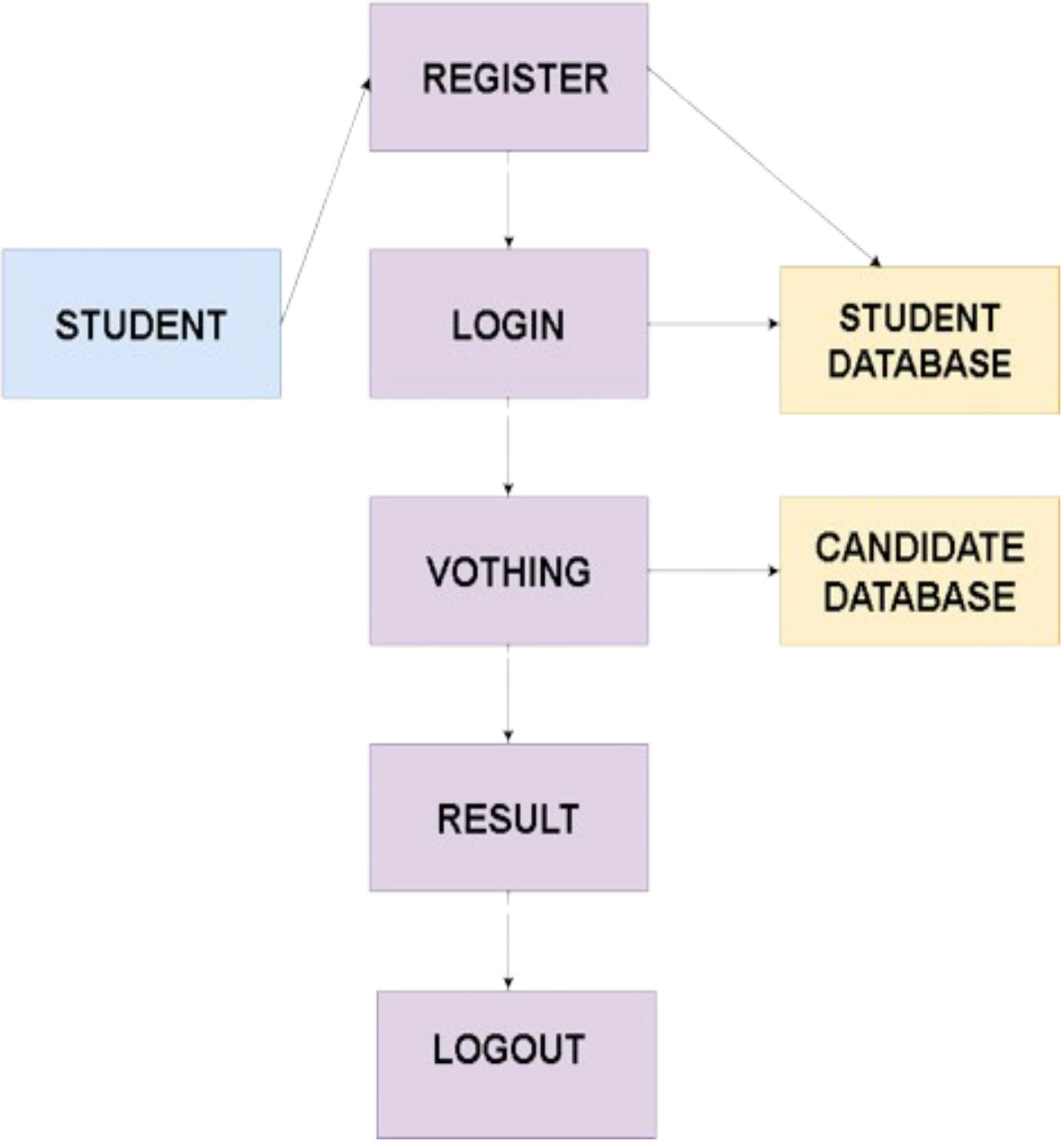


3.5 Data Flow Diagram

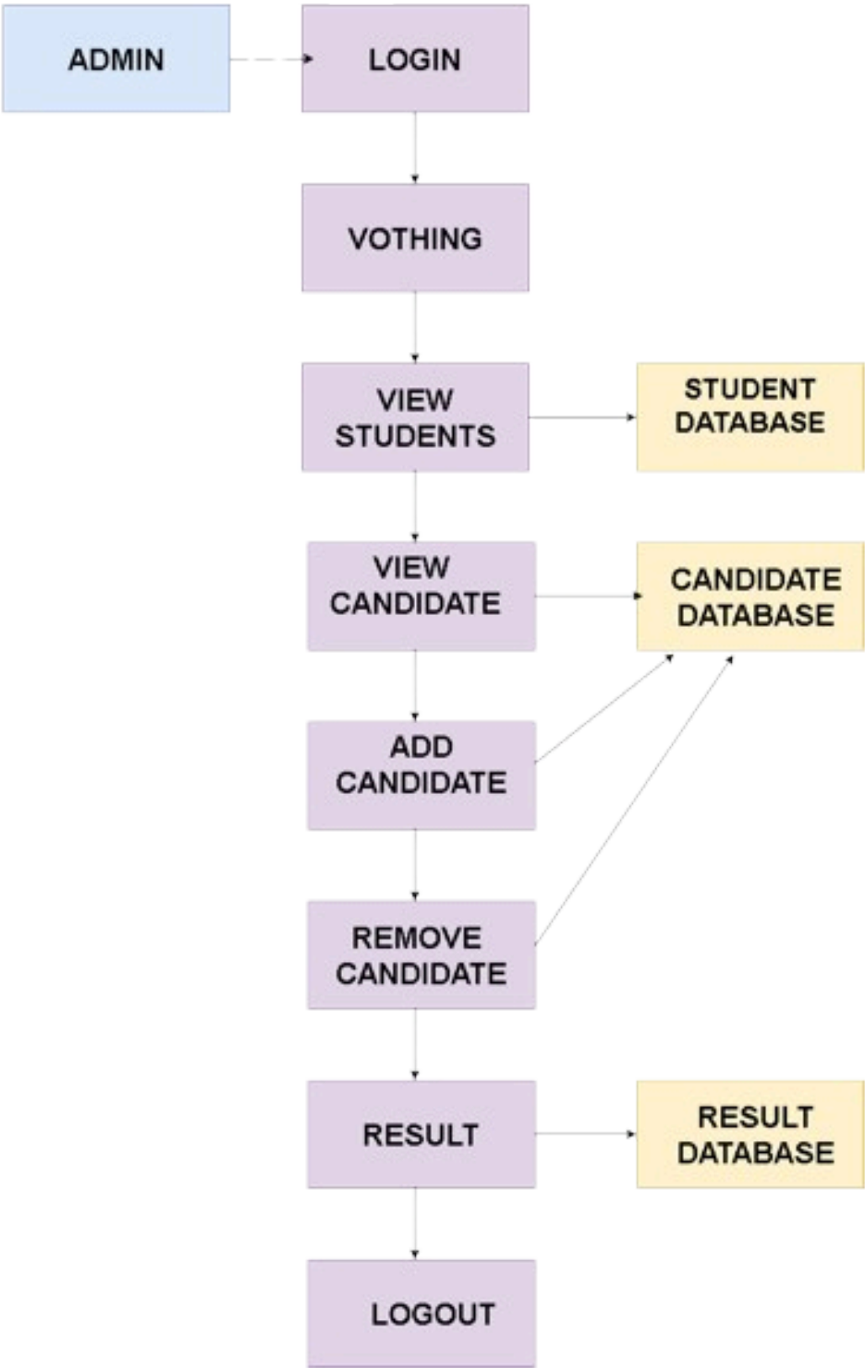
LEVEL 0



LEVEL 1


REGISTER
STUDENT
LOGIN
STUDENT DATABASE
VOTHING
CANDIDATE DATABASE
RESULT
LOGOUT

LEVEL 2


ADMIN
LOGIN
VOTHING
VIEW STUDENTS
STUDENT DATABASE
VIEW CANDIDATE
CANDIDATE DATABASE
ADD CANDIDATE
REMOVE CANDIDATE
RESULT
RESULT DATABASE
LOGOUT

3.6 Software Requirement Specification

1. Purpose:

The purpose of the College Voting System is to provide a secure and user-friendly platform for conducting elections within a college environment. The system aims to eliminate the traditional paper-based voting process by introducing an automated online platform that ensures accuracy, transparency, and efficiency. The system will allow students to securely log in, cast their votes, and view real-time results. It will also enable administrators to manage candidates, monitor voting activities, and generate accurate election results.

2. Scope:

The College Voting System is designed to automate and simplify the voting process within a college. It will ensure a secure and transparent election process by allowing students to vote online through an authenticated system. The system will handle the following tasks:

- ✔ User registration and authentication
- ✔ Candidate registration and display
- ✔ Secure online voting
- ✔ Real-time vote counting and result generation
- ✔ Administrator control for managing users, candidates, and election data

3. Functional Requirements:

Functional requirements define the specific behavior and functions of the College Voting System:

<u>User Registration and Authentication:</u>

- Students should be able to register with valid credentials.

- Users should be authenticated using a secure login process.
- The system should prevent multiple registrations from the same user

Candidate Management:

- Administrators should be able to add, update, and delete candida profiles.
- Candidate details should be displayed to voters befotere voting begins.

Voting Process:

- Students should be able to select a candidate and cast their vote.
- The system should prevent multiple voting by the same user.
- Votes should be encrypted and securely stored in the database.

Result Generation:

- The system should count votes in real time.
- Results should be automatically calculated and displayed after the voting period ends.
- Only authorized administrators should be able to reset or clear election data.

Administrator Functions:

- Admins should be able to monitor and control the election process.
- Admins should be able to modify candidate details and monitor voting status.
- Admins should have access to detailed voting logs and reports.

4. Non-Functional Requirements:

Non-functional requirements define the quality attributes and performance of the system:

Performance:

- The system should be able to handle up to 500 concurrent users without performance issues.
- Voting results should be calculated and displayed within 5 seconds after voting ends.

Security:

- User data and voting records should be encrypted using AES encryption.
- The system should have protection against SQL injection and cross-site scripting (XSS) attacks.
- Unauthorized access to the system should be restricted.

Usability:

- The system should have an intuitive and responsive user interface.
- The interface should work on desktops, tablets, and mobile devices.

Availability:

- The system should have 99.9% uptime during the voting period.
- In case of server failure, the system should recover within 10 minutes.

Maintainability:

- The codebase should follow industry standards to allow easy updates and maintenance.
- Logs should be generated and stored for future auditing and debugging.

4 . SYSTEM DESIGN

4.1

SYSTEM DESIGN

Module	Functionality	Input	Output
Admin	- Manage candidate registrations - Approve or reject candidate applications - View and manage voter lists - Monitor and finalize election results	- Admin login credentials - Candidate details - Voter details	- Candidate approval/rejection confirmation - Vote r list management - Election results
Voter	- Login to the voting system - Cast a vote for candidates - View election results after voting ends	-- Voter login credentials - Selected candidate for each	- Voting confirmation - Election results (after voting ends)
Candidate	- Register for elections - View vote count (if allowed)	- Candidate registration details - Login credentials	- Registration approval/rejection - Vote count status (if applicable)

4.2

Process design

The process design for the online college voting system involves structuring the workflow for conducting fair and transparent elections within the college. The process begins with candidate registration, where eligible students submit their nominations for approval by the Admin. Once approved, candidates' details are added to the system for public viewing. User authentication ensures only registered students and staff can log in and participate in the voting process. On the election day, voters cast their votes online, with the system verifying each vote to prevent duplication. The vote counting and result computation occur automatically, ensuring accuracy and transparency. Finally, the Admin reviews and publishes the election results, notifying all stakeholders. Throughout the process, security measures, real-time updates, and role-based access controls are enforced to ensure a smooth and fair voting experience.

1. User Management Module:

Process Design:

- **User Registration:**
    - New users (Students) register by providing details (name, register number, department, year, etc.).
    - System verifies information and generates unique login credentials.
- **User Authentication:**
    - Users log in using credentials (register no and password).
    - System authenticates users against stored records and grants access based on roles.

2. Candidate Management Module:

Process Design:

- **Candidate Registration:**
    - Eligible students submit nominations by providing required details (name, image, department, category etc.).
    - Admin reviews and approves/rejects candidates based on eligibility criteria.
- **Candidate Approval & Listing:**
    - Approved candidates are displayed in the system for voter review.

3. Voting Process Module:

Process Design:

- **Voter Authentication:**
    - Students log in and verify their eligibility before casting a vote.
    - System ensures each user can only vote once.
- **Vote Casting:**
    - Users select their preferred candidate and confirm their vote.
    - Votes are encrypted and securely stored in the database.
- **Vote Submission & Validation:**
    - System records the vote and prevents duplicate voting.
    - Real-time updates ensure vote integrity and transparency.

4. Vote Counting & Result Processing Module:

Process Design:

- **Automated Vote Counting:**
  - The system automatically tallies votes after the election ends.
  - Results are calculated based on secure and transparent algorithms.
- **Result Verification:**
  - Admin verifies the computed results before publishing.
  - Any discrepancies are flagged and reviewed.
- **Result Announcement:**
  - Verified results are published on the system.
  - Notifications are sent to candidates and voters.

5. Admin Management Module:

Process Design:

- **User & Candidate Management:**
  - Admin manages user registrations, candidate approvals, and voting rules.
- **Election Setup:**
  - Admin configures election dates, positions, and voting eligibility.
- **System Monitoring:**
  - Admin ensures security, prevents fraud, and resolves system issues

4.3 Table Design

Tables

- Students
- Candidate
- Result

1. Students_registration

Purpose: to store students details

Primary key: id

Field Name	Data type	Length	Description
id	INT	------	Primary key, Auto-increment, Unique ID for each student
name	VARCHAR	100	Full name of the student
reg_namber	VARCHAR	50	Unique registration number for the student
date_of_birth	DATE	-----	Student's date of birth
gender	VARCHAR	10	Gender of the student (Male/Female/Other)
email	VARCHAR	100	Student's email address
department	VARCHAR	50	Department of the student
year_of_study	INT	----	Current year of study
password	VARCHAR	255	password

2. candidates

Purpose: to store candidate details

Primary key:

Field Name	Data type	Length	Description
id	INT	—	Primary key, unique identifier
name	VARCHAR	50	Candidate's name
image	VARCHAR	50	Candidate's image path
department	ENUM	—	Candidate's department
year	ENUM	—	Candidate's academic year
category	ENUM	—	Candidate category (General/Reserved)

3. RESULT&VOTE

Purpose: To store vote count result

Primary key: id

	Field Name	Data Type	Length	Description
1	id	int	N/A	Primary key, unique identifier
2	user_id	varchar	5	Unique user identifier
3	page	varchar	50	Page name or section where vote is recorded
4	general_vote	varchar	50	Stores the candidate ID for general category
5	reserved_vote	varchar	50	Stores the candidate ID for reserved category
6	timestamp	timestamp	N/A	Records the time when the vote was cast

5 TESTING AND IMPLEMENTATION

TESTING AND IMPLEMENTATION

To ensure the successful testing and implementation of the college voting system with the specified modules and functionalities, a systematic approach is necessary. Firstly, the system's development should follow a well-defined plan, starting with database design. This includes creating tables for user information,candidates information, result details.Backend development comes next, utilizing PHP for server-side scripting and MySQL for database management to handle authentication, vote processing, and result computation efficientlyAPI endpoints need to be defined in PHP for user authentication, candidate management, and result processing, ensuring seamless communication between the frontend and the MySQL database. Simultaneously, frontend development should proceed, designing user interfaces for each module using HTML, CSS, JAVA SCRIPT to provide a smooth voting experience.

The authentication mechanisms should be thoroughly tested to ensure secure login and proper access control based on user roles and permissions. At this stage, unit testing is essential, with tests written for each module's functionality using appropriate PHP testing frameworks like PHPUnit. These tests cover authentication, data retrieval from the MySQL database, and request handling, ensuring that each component functions correctly in isolation.

Integration testing follows, where interactions between different modules are tested to ensure seamless communication and data flow between the PHP backend and the MySQL database. Key scenarios, such as user login, candidate registration, vote submission, and result computation, are thoroughly tested to validate system behavior under various conditions. User Acceptance Testing (UAT) is then conducted, involving stakeholders such as the Principal, HOD, Staff, and Students to verify that the system meets their requirements. The frontend, developed using HTML and CSS, JAVA SCRIPT is assessed for usability to ensure an intuitive and user-friendly voting experience.

In summary, a structured approach covering database design, backend and frontend development, comprehensive testing (unit, integration, and UAT), and deployment with continuous monitoring and maintenance is crucial for the successful implementation of the college voting system. This ensures the system operates securely, meets user requirements, and provides a smooth and transparent voting experience for all stakeholders, including students, faculty, and administrators.

5.1 Testing Techniques

Unit Testing:

- Test each module separately to ensure expected functionality.
- Validate user authentication, vote casting, candidate registration, and result computation.
- Ensure correct data retrieval and error handling in MySQL database.
- Use PHPunit for backend testing.

Integration Testing:

Integration testing ensures smooth communication between frontend, backend (PHP), and database (MySQL). It verifies key functions like user login, candidate registration, vote submission, and result display. The system must process votes correctly, update the database in real time, and enforce role-based access control. Error handling and system recovery are tested for unexpected failures. This ensures a secure, seamless, and efficient voting process.

User Acceptance Testing (UAT):

UAT involves Principal, HOD, Staff, and Students to validate system functionality and usability. The Admin manages candidate and user details to ensure eligibility. Key features like user authentication, vote casting, and result computation are tested. UAT sessions check system security, transparency, and ease of use. Feedback from users helps refine the system for better performance. This ensures a reliable and user-friendly voting experience.

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6 Conclusion

In conclusion, the development, testing, and implementation of the college voting system require a structured and comprehensive approach to ensure its efficiency, security, and usability for all stakeholders, including the Principal, HOD, Staff, and Students. By systematically following key steps—from database design and backend development using PHP and MySQL to frontend implementation with HTML, CSS, and Bootstrap—the system can be developed with precision and reliability.

The integration of robust authentication mechanisms ensures secure access control, while unit and integration testing validate the system's functionality and seamless interaction between different modules. User Acceptance Testing (UAT) plays a crucial role in refining the system based on feedback from end-users, ensuring an intuitive and user-friendly voting experience. Once deployed in a secure and reliable hosting environment, continuous monitoring and maintenance further guarantee the system's stability and responsiveness in real-world elections.

Ultimately, the successful implementation of the college voting system depends on meticulous planning, rigorous testing, and a commitment to meeting the diverse needs of users. By leveraging technology to enhance transparency, security, and accessibility, the system serves as a valuable tool in streamlining the voting process, fostering democratic participation, and ensuring fair elections within the college community.

7.SCOPE FOR FUTURE DEVELOPMENT

The college voting system described holds significant potential for future development, offering opportunities to enhance security, accessibility, and user experience to better serve the evolving needs of educational institutions. One key area for future improvement is the integration of advanced analytics and data-driven insights. By incorporating data analytics, the system can analyze past voting patterns, voter turnout rates, and candidate performance to improve election processes, anticipate trends, and ensure fair and transparent voting.

Furthermore, adding features such as automated voter registration and secure ballot verification can streamline election management, reducing manual efforts and minimizing errors. Integration with communication platforms can facilitate real-time notifications and updates, ensuring seamless coordination among stakeholders and enhancing voter participation.

Another area for development involves leveraging emerging technologies such as artificial intelligence (AI) and machine learning (ML). AI-driven chatbots can be integrated to assist voters by answering queries and guiding them through the voting process, while ML algorithms can analyze election data to detect anomalies and prevent fraudulent activities, ensuring a secure and fair voting environment.

Additionally, enhancing the system's mobile capabilities can increase accessibility, allowing students and faculty to vote conveniently from their devices. A mobile-friendly platform or dedicated app can include features such as voter authentication, live election updates, and result announcements, improving engagement and participation.

Moreover, incorporating support for remote and hybrid voting can address the needs of off-campus students and faculty, ensuring inclusivity. Integration with secure voting platforms and identity verification mechanisms can further enhance the credibility and reliability of the system. Ensuring compliance with data privacy regulations and implementing robust security measures, such as encryption and multi-factor authentication, will be essential to protect voter information and maintain trust.

Overall, the future development of the college voting system presents numerous opportunities for innovation and efficiency, driven by advancements in technology, evolving user expectations, and the growing importance of secure digital voting. By emerging technologies, enhancing security protocols, and focusing on user-centric design, the system can continue to evolve and adapt to meet the needs of colleges and universities in the digital age.

8 .APPENDIX

APPENDIX

- System source code
- GUI screenshots
- System minimum requirements

Source Code

❖ login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Voting Login</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <style>
    body {
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #23232e, #030509);
      color: white;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      overflow: hidden;
      position: relative;
    }

    /* Background Text (College Name) */
    body::before {
      content: "COLLAGE VOTING SYSTEM "; /* Change to your college name */
      position: absolute;
      top: 50%;
      left: 50%;
      transform: translate(-50%, -50%);
      font-size: 100px;
      font-weight: bold;
      color: rgba(248, 244, 244, 0.05); /* Low opacity */
      z-index: 0;
      white-space:
      nowrap;
      pointer-events: none;
    }
```

```
.login-container {
   background: rgba(255, 255, 255, 0.1);
   padding: 40px;
   border-radius: 10px;
   box-shadow: 0px 0px 15px rgba(255, 255, 255,
   0.2); text-align: center;
   width: 400px;
   animation: fadeIn 1s ease-in-out;
   position: relative;
   z-index: 1;
}

.login-container h2 {
   margin-bottom:
   25px;
}

.form-control {
   background: rgba(255, 255, 255, 0.2);
   border: none;
   color: white;
   padding: 12px;
   font-size: 16px;
}

.form-control::placeholder {
   color: rgba(255, 255, 255,
   0.7);
}

.btn-login, .btn-register
   { font-weight: bold;
   transition: transform 0.3s, background-color 0.3s;
   padding: 12px;
   font-size: 18px;
}

.btn-login {
   background-color: rgb(1, 1, 10);
   color: white;
}
.btn-login:hover {
background-color: navy;
transform: scale(1.05);
```

```
        }

        .btn-register {
            background-color: rgb(1, 1, 10);
            color: white;
            margin-top: 12px;
        }

        .btn-register:hover {
            background-color:
            green; transform:
            scale(1.05);
        }

        .fade-in {
            opacity: 0;
            transform: translateY(20px);
            animation: fadeInUp 0.8s ease-in-out forwards;
        }

        @keyframes fadeIn {
            0% { opacity: 0; }
            100% { opacity: 1; }
        }

        @keyframes fadeInUp {
            0% { opacity: 0; transform: translateY(20px); }
            100% { opacity: 1; transform: translateY(0); }
        }
    </style>
  </head>
  <body>
  <div class="login-container">
    <h2 class="fade-in">College Voting Login</h2>
    <form action="login.php" method="POST">
        <!-- Registration Number Input -->
        <div class="mb-3 fade-in" style="animation-delay: 0.2s;">
<input type="text" class="form-control" name="reg_number" placeholder="Registration Number"
  required>
        </div>
        <!-- Password Input -->
        <div class="mb-3 fade-in" style="animation-delay: 0.4s;">
            <input type="password" class="form-control" name="password" placeholder="Password" required>
```

```
    </div>
    <!-- Login Button -->
                <button type="submit" class="btn btn-login w-100 fade-in" style="animation-delay:
0.6s;">Login</button>
  </form>
  <!-- Register Button -->
              <button class="btn btn-register w-100 fade-in" style="animation-delay: 0.8s;"
onclick="window.location.href='register.html'">Register</button>
</div>

</body>
</html>
```

❖ register.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Registration</title>
              <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <style>
    body {
      background-color:
      #000000; display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      color: white;
      padding:
      10px;
    }
    .registration-container
      { background:
      #ffffff; padding:
      20px; border-radius:
      10px;
      border: 3px solid black;
      box-shadow: 0 4px 8px rgba(255, 255, 255, 0.2);
      width: 450px;
      color: black;
    }
    .btn-primary {
      background-color: #3D73FF;
```

```
                    border: none;
            width: 100%;
font-size:
16px; padding: 8px;

                }
                .btn-primary:hover {
                    background-color:
                    #2c5dd9;
                }
                .form-control {
                    margin-bottom: 12px;
                    border: 1px solid black;
                    font-size: 14px;
                    padding: 8px;
                }
                h3 {
                    color: #000000;
                    font-weight:
                    bold; font-size:
                    22px; text-align:
                    center;
                    margin-bottom: 15px;
                }
                label {
                    font-size: 15px;
                    font-weight:
                    bold;
                }
            </style>
        </head>
        <body>

        <div class="registration-container">
            <h3>Registration</h3>
            <form id="registrationForm" action="register.php" method="POST">
                <label class="form-label">Name</label>
                <input type="text" class="form-control" name="name" placeholder="Enter full name" required>
                <label class="form-label">Registration Number</label>
                    <input type="text" class="form-control" name="registration_number" placeholder="Enter
        registration number" required>
                <label class="form-label">Date of Birth</label>
                <input type="date" class="form-control" name="dob" required>
```

```
        <label class="form-label">Gender</label>
        <select class="form-control" name="gender" required>

          <option value="" disabled selected>Select gender</option>
          <option value="male">Male</option>
          <option value="female">Female</option>
          <option value="other">Other</option>
        </select>

        <label class="form-label">Email</label>
           <input type="email" class="form-control" name="email" placeholder="Enter your Email Id"
      required>

        <label class="form-label">Department</label>
        <select class="form-control" name="department" required>
          <option value="" disabled selected>Select department</option>
          <option value="bca">BCA</option>
          <option value="economics">Economics</option>
          <option value="bba">BBA</option>
          <option value="ai">AI</option>
          <option value="bca_ai">BCOM</option>
        </select>

        <label class="form-label">Year of Study</label>
        <select class="form-control" name="year_of_study" required>
          <option value="" disabled selected>Select year</option>
          <option value="1">1st Year</option>
          <option value="2">2nd Year</option>
          <option value="3">3rd Year</option>
          <option value="4">4th Year</option>
        </select>

        <label class="form-label">Set Password</label>
         <input type="password" class="form-control" name="password" placeholder="Enter
      password" required>

        <label class="form-label">Confirm Password</label>
        <input type="password" class="form-control" name="confirm_password" placeholder="Confirm
      password" required>

        <button type="submit" class="btn btn-primary mt-3">SUBMIT</button>
        </form>
      </div>
      </body>
</html>
```

❖ connect.php

```php
<?php
$host = 'localhost';
$db = 'user_db';
$user = 'root';
$pass = '';
$charset = 'utf8mb4';
$dsn = "mysql:host=$host;dbname=$db;charset=$charset";
try {
    $conn = new PDO($dsn, $user, $pass);
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
} catch (PDOException $e) {
    die("Connection failed: " . $e->getMessage());
}
```

❖ register.php

```php
<?php
require 'connect.php';

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $name = $_POST['name'];
    $registration_number = $_POST['registration_number'];
    $dob = $_POST['dob'];
    $gender = $_POST['gender'];
    $email = $_POST['email'];
    $department = $_POST['department'];
    $year_of_study = $_POST['year_of_study'];
    $password = $_POST['password'];
    $confirm_password = $_POST['confirm_password'];

    // 🔒 Password Match Check
    if ($password !== $confirm_password) {
        echo "<script>alert('Passwords do not match!'); window.history.back();</script>";
        exit();
    }

    // 🔒 Password Hashing
        $hashed_password = password_hash($password, PASSWORD_DEFAULT); try {
```

```
        $check = $conn->prepare("SELECT * FROM students WHERE email = :email OR reg_number
= :reg_number");
        $check->bindParam(':email', $email, PDO::PARAM_STR);
        $check->bindParam(':reg_number', $registration_number, PDO::PARAM_STR);
        $check->execute();


        if ($check->rowCount() > 0) {
                                echo "<script>alert('Email or Registration Number already exists!');
window.history.back();</script>";
            exit();
        }


        $stmt = $conn->prepare("INSERT INTO students
            (name, reg_number, date_of_birth, gender, email, department, year_of_study, password)
            VALUES
            (:name, :reg_number, :dob, :gender, :email, :department, :year_of_study, :password)");

        $stmt->bindParam(':name', $name);
        $stmt->bindParam(':reg_number', $registration_number);
        $stmt->bindParam(':dob', $dob);
        $stmt->bindParam(':gender', $gender);
        $stmt->bindParam(':email', $email);
        $stmt->bindParam(':department', $department);
        $stmt->bindParam(':year_of_study', $year_of_study);
        $stmt->bindParam(':password', $hashed_password);

        $stmt->execute();


            echo "<script>alert('Registration successful! Redirecting to login...'); window.location.href =
'login.html';</script>";
        exit();

    } catch (PDOException $e) {
        echo "<script>alert('Error: " . $e->getMessage() . "'); window.history.back();</script>";
    }
}

?>
```

❖ **login.php**

```
<?php session_start();
require 'connect.php';

    ini_set('display_errors',
    1);
    error_reporting(E_ALL);

    if ($_SERVER["REQUEST_METHOD"] == "POST") {
      $reg_number = htmlspecialchars($_POST['reg_number']);
      $password = htmlspecialchars($_POST['password']);

      try {

        if ($reg_number === 'admin123' && $password === 'admin@123') {
          $_SESSION['admin'] =
          $reg_number; header("Location:
          login.html"); exit();
        }


        $stmt = $conn->prepare("SELECT * FROM students WHERE reg_number = :reg_number");
        $stmt->bindValue(':reg_number', $reg_number);
        $stmt->execute();
        $user = $stmt->fetch(PDO::FETCH_ASSOC);
        if ($user) {
          // Debugging output:
          echo "User found: " . $user['reg_number'];

          if (password_verify($password, $user['password'])) {
            $_SESSION['user'] = $user['reg_number'];
            header("Location: home.html");
            exit();
          } else {
            echo "<script>alert('Incorrect Password'); window.history.back();</script>";
          }
        } else {
          echo "<script>alert('Invalid Register Number'); window.history.back();</script>";
        }
      } catch (PDOException $e) {
        echo "Error: " . $e->getMessage()
```

home.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Voting System</title>
            <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <style>
    body {
      font-family: Arial,
      sans-serif;
      background-color: #f8f9fa;
    }
    .navbar {
      background-color: #f9fbfb;
    }
    .navbar-brand, .nav-link {
      color: black !important;
    }
    .navbar-brand img {
      height: 50px;
      margin-right:
      10px;
    }
    .hero {
      background: linear-gradient(-45deg, #121313, #1c1c1c, #121313, #000000);
      background-size: 400% 400%;
      animation: animateBackground 5s ease infinite;
      color: white;
      text-align: center;
      padding: 80px 20px;
    }
    @keyframes animateBackground {
      0% { background-position: 0% 50%; }
      50% { background-position: 100% 50%; }
      100% { background-position: 0% 50%; }
    }
    .features {
      padding: 40px 20px;
    }
    .feature-box {
      background:
      white;
```

```
            padding: 60px;
            border-radius:
            8px;
            box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
            opacity: 0;
            transform: translateY(20px);
            animation: fadeInUp 1s ease-in-out forwards;
        }
        .feature-box:nth-child(1) { animation-delay: 0.3s; }
        .feature-box:nth-child(2) { animation-delay: 0.6s; }
        .feature-box:nth-child(3) { animation-delay: 0.9s; }
        @keyframes fadeInUp {
            0% { opacity: 0; transform: translateY(20px); }
            100% { opacity: 1; transform: translateY(0); }
        }
        .about {
            background-color:
            #ddd; padding: 40px
            20px;
        }
        .footer {
            background-color:
            #222; color: white;
            text-align: center;
            padding: 20px 0;
            margin-top:
            20px;
        }
    </style>
</head>
<body>

    <!-- Navbar -->
    <nav class="navbar navbar-expand-lg">
        <div class="container">
            <a class="navbar-brand fw-bold" href="home.html" style="color: #000; font-size: 1.8rem;
letter-spacing: 2px;">
            College Voting
            </a>
<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
            <span class="navbar-toggler-icon"></span>
            </button action="logout.php" method="post">
```

```
  <div class="collapse navbar-collapse" id="navbarNav">
<ul class="navbar-nav ms-auto"

<li class="nav-item"><a class="nav-link" href="adminlogin.html" style="color: black
            !important;">Admin</a></li>
                                <li class="nav-item"><a class="nav-link" href="about.html" style="color: black
            !important;">About</a></li>
                                <li class="nav-item"><a class="nav-link" href="results.html" style="color: black
            !important;">Results</a></li>
                                 <li class="nav-item"><a class="nav-link" href="help.html" style="color: black
            !important;">Help</a></li>
                                <li class="nav-item"><a class="nav-link" href="logout.php" style="color: black
            !important;">Logout</a></li>
                    </ul>
                  </div>
                </div>
              </nav>
              <!-- Hero Section -->
              <div class="hero">
                <h1>Welcome to the College Election</h1>
                <p>Secure & Transparent Online Voting for College Elections</p>
                <a href="voting.html" class="btn btn-light mt-3">Voting</a>
              </div>

              <!-- Features Section -->
              <div class="container features text-center">
                <h2>Why Choose Our Voting System?</h2>
                <div class="row mt-4">
                  <div class="col-md-4">
                    <div class="feature-box">
                      <h4>Secure Voting</h4>
                      <p>End-to-end encrypted voting system ensuring fair elections.</p>
                    </div>
                  </div>
                  <div class="col-md-4">
                    <div class="feature-box">
                      <h4>Real-Time Results</h4>
                      <p>Instant and transparent results without any manual counting.</p>
                    </div>
                  </div>
                  <div class="col-md-4">
                    <div class="feature-box">
                      <h4>Easy Registration</h4>
                      <p>Register easily using your student ID & Email.</p>
                    </div>
```

```
<li class="nav-item"><a class="nav-link" href="adminlogin.html" style="color: black
!important;">Admin</a></li>
                    <li class="nav-item"><a class="nav-link" href="about.html" style="color: black
!important;">About</a></li>
                    <li class="nav-item"><a class="nav-link" href="results.html" style="color: black
!important;">Results</a></li>
                     <li class="nav-item"><a class="nav-link" href="help.html" style="color: black
!important;">Help</a></li>
                    <li class="nav-item"><a class="nav-link" href="logout.php" style="color: black
!important;">Logout</a></li>
            </ul>
          </div>
        </div>
      </nav>

      <!-- Hero Section -->
      <div class="hero">
        <h1>Welcome to the College Election</h1>
        <p>Secure & Transparent Online Voting for College Elections</p>
        <a href="voting.html" class="btn btn-light mt-3">Voting</a>
      </div>

      <!-- Features Section -->
      <div class="container features text-center">
        <h2>Why Choose Our Voting System?</h2>
        <div class="row mt-4">
          <div class="col-md-4">
            <div class="feature-box">
              <h4>Secure Voting</h4>
              <p>End-to-end encrypted voting system ensuring fair elections.</p>
            </div>
          </div>
          <div class="col-md-4">
            <div class="feature-box">
              <h4>Real-Time Results</h4>
              <p>Instant and transparent results without any manual counting.</p>
            </div>
          </div>
          <div class="col-md-4">
            <div class="feature-box">
              <h4>Easy Registration</h4>
              <p>Register easily using your student ID & Email.</p>
            </div>
          </div>
```

```
        </div>
    </div>
    <!-- About Section -->
    <div class="about text-center">
        <h2>About Our Voting System</h2>
        <p>
            Our college voting system ensures a secure and fair electoral process, allowing students to vote for their representatives in a hassle-free online environment.
        </p>
    </div>

    <!-- Footer -->
    <div class="footer">
        <p>&copy; 2025 College Voting System | Contact: siasindia@edu.ins</p>
    </div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

❖ **admincandidate.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Candidate Admin Page</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
    <style>
        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(135deg, #23232e, #030509);
            color: white;
```

```
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    margin: 0;
    overflow: hidden;
}

.container {
    background: rgba(255, 255, 255, 0.1);
    padding: 30px;
    border-radius: 12px;
    box-shadow: 0px 0px 15px rgba(255,
255, 255, 0.2);
    width: 400px;
    text-align: center;
    animation: fadeIn 0.8s ease-in-out;
}

h1 {
    font-size: 24px;
    margin-bottom: 20px;
}

.input-group {
    margin-bottom: 15px;
    text-align: left;
}

label {
    font-weight: bold;
    display: block;
    margin-bottom: 5px;
    color: rgba(255, 255, 255, 0.8);
}
```

```
input, select {
    width: 100%;
    padding: 12px;
    background: rgba(255, 255, 255, 0.2);
    border: none;
    border-radius: 8px;
    color: white;
    box-sizing: border-box;
    transition: background-color 0.3s ease;
    font-size: 16px;
}

input:focus, select:focus {
    background-color: #555;
    outline: none;
}

.submit-btn {
    background-color: rgb(1, 1, 10);
    color: white;
    border: none;
    padding: 14px;
    font-size: 18px;
    width: 100%;
    border-radius: 8px;
    cursor: pointer;
    font-weight: bold;
    transition: background-color 0.3s ease,
transform 0.2s;
    margin-top: 10px;
}

.submit-btn:hover {
    background-color: navy;
    transform: scale(1.05);
}
```

```
        @keyframes fadeIn {
            0% { opacity: 0; }
            100% { opacity: 1; }
        }
    </style>
</head>
<body>

<div class="container">
    <h1>Candidate Registration</h1>

    <form action="admin_candidate.php"
method="POST"
enctype="multipart/form-data">
        <div class="input-group">
            <label>Name:</label>
            <input type="text" name="name"
placeholder="Enter name" required>
        </div>

        <div class="input-group">
            <label>Image:</label>
            <input type="file" name="image"
required>
        </div>

        <div class="input-group">
            <label>Department:</label>
            <select name="department" required>
                <option
value="BCOM">BCOM</option>
                <option
value="BBA">BBA</option>
                <option
value="BCA">BCA</option>
                <option value="AI">AI</option>
```

```
            <option value="BA ENGLISH">BA
ENGLISH</option>
            <option value="BA
ENGLISH">ECONOMICS</option>
        </select>
    </div>

    <div class="input-group">
        <label>Year:</label>
        <select name="year" required>
            <option value="First Year">First
Year</option>
            <option value="Second Year">Second
Year</option>
            <option value="Third Year">Third
Year</option>
            <option value="Final Year">Final
Year</option>
        </select>
    </div>

    <div class="input-group">
        <label>Category:</label>
        <select name="category" required>
            <option
value="General">General</option>
            <option
value="Reserved">Reserved</option>
        </select>
    </div>

    <button type="submit"
class="submit-btn">Submit</button>
  </form>
</div>
</body>
</html>
```

admin_candidate.php

```
<?php
// Database connection
$servername = "localhost";
$username = "root"; // Default for XAMPP
$password = "";     // Default for XAMPP (empty)
$database = "user_db";

$conn = new mysqli($servername, $username, $password, $database);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
// Handling form submissio

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $name = $_POST['name'];
    $department = $_POST['department'];
    $year = $_POST['year'];
    $category = $_POST['category'];
    // Handling image upload
    $target_dir = "uploads/";
    if (!file_exists($target_dir)) {
        mkdir($target_dir, 0777,
        true);
    }
    $image = $target_dir . basename($_FILES["image"]["name"]);
    move_uploaded_file($_FILES["image"]["tmp_name"],
    $image);
    // Insert into database
    $stmt = $conn->prepare("INSERT INTO candidates (name, image, department, year, category) VALUES (?,
?, ?, ?, ?)");
    $stmt->bind_param("sssss", $name, $image, $department, $year, $category);

    if ($stmt->execute()) {
        echo "<script>alert('Candidate Registered Successfully!');
window.location.href='admin_candidate.html';</script>";
    } else {
        echo "Error: " . $stmt->error;
    }
    // Close statement and connection
    $stmt->close();
    $conn->close();
}
?>
```

voting.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Voting - Select Department</title>
            <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <style>
    body {
      font-family: Arial, sans-serif;


      background-color: #111112;
      color: white;
    }
    .container {
      text-align: center;
      margin-top:
      50px;
    }
    .department {
      cursor: pointer;
      transition: transform 0.3s ease-in-out;
      padding: 20px;
    }
    .department:hover {
      transform:
      scale(1.08);
    }
    .department img {
      width: 100%;
      max-width: 250px; /* Increased Image Size
      */ border-radius: 10px;
      background: rgba(255, 255, 255, 0.1); /* Light shadow background */
      padding: 10px;
      box-shadow: 0 8px 16px rgba(255, 255, 255, 0.3);
    }
    .department a {
      text-decoration: none; /* Removes underline */
      color: white; /* Text changed to white */
    }
    .department h4 {
      margin-top:
      10px; font-size:
      20px;
```

```
        }
        .footer {
            background-color:
            #222; color: white;
            text-align: center;
            padding: 20px 0;
            margin-top:
            20px;
        }
    </style>
</head>
<body>

    <div class="container">
        <h2> DEPARTMENT</h2>

        <p>Select Your Department</p>


        <!-- First Row -->
        <div class="row justify-content-center">
            <div class="col-md-4 department">
                <a href="passkey/passbba.html">
                    <img src="bba.png" alt="BBA Department">
                    <h4>BBA</h4>
                </a>
            </div>
            <div class="col-md-4 department">
                <a href="passkey/passbca.html">
                    <img src="bca.png" alt="BCA Department">
                    <h4>BCA</h4>
                </a>
            </div>
            <div class="col-md-4 department">
                <a href="passkey/passai.html">
                    <img src="ai.png" alt="AI Department">
                    <h4>AI</h4>
                </a>
            </div>
        </div>
```

```
        </div>

        <!-- Second Row -->
        <div class="row justify-content-center">
          <div class="col-md-4 department">
            <a href="passkey/passbcom.html">
              <img src="bcom.png" alt="BCom Department">
              <h4>BCom</h4>
            </a>
          </div>
          <div class="col-md-4 department">
            <a href="passkey/passeng.html">
              <img src="eng.png" alt="BA English Department">
              <h4>BA English</h4>
            </a>
          </div>
          <div class="col-md-4 department">
            <a href="passkey/passeco.html">
              <img src="eco.png" alt="Economics Department">
              <h4>Economics</h4>
            </a></div>
        </div>
      </div>

      <div class="footer">
        <p>&copy; 2025 College Voting System | Contact: siasindia@edu.ins</p>
      </div>

    </body>
    </html>
```

❖ **passkey.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Department Voting - Year Selection</title>
                <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
   <style>
      body {
         font-family: Arial,
         sans-serif;
         background-color: #121212;
         color: white;
      }
      .container {
         max-width: 600px;
         margin: 100px auto;
         background: rgba(255, 255, 255, 0.1);
         padding: 40px;
         border-radius: 15px;
         box-shadow: 0 12px 24px rgba(255, 255, 255,
         0.2); text-align: center;
      }
      select, input {
         width: 100%;
         padding:
         14px; border:
         none;
         border-radius:
         8px; font-size:
         18px;
      }
      select {
         margin-bottom: 20px;

      }
      input {
         background: #fff;
      }
```

```
 .btn-submit {
background: #007bff; color: white;
                border: none;
                padding: 14px;
                width: 100%;
                margin-top: 20px;
                border-radius:
                8px; font-size:
                20px; cursor:
                pointer; transition:
                0.3s;
              }
              .btn-submit:hover {
                background:
                #0056b3;
              }
              .error {
                color: red;
                margin-top:
                10px; font-size:
                16px;
              }
            </style>
          </head>
          <body>

            <div class="container">
              <h2>BCA</h2>
              <p>Select your year and enter the passkey to proceed.</p>

              <form id="votingForm">
                <label for="year">Select Year:</label>
                <select id="year" required>
                  <option value="" disabled selected>-- Choose Year --</option>
                  <option value="1st">1st Year</option>
                  <option value="2nd">2nd Year</option>
                  <option value="3rd">3rd Year</option>
                  <option value="4th">4th Year</option>
                </select>

                <label for="passkey">Enter Passkey:</label>
                <input type="password" id="passkey" placeholder="Enter department passkey" required>
```

```
            <p class="error" id="errorMessage"></p>

            <button type="submit" class="btn-submit">Proceed to Voting</button>
        </form>
    </div>

    <script>
        document.getElementById("votingForm").addEventListener("submit", function(event) {
            event.preventDefault();

            let passkey = document.getElementById("passkey").value;
            let year = document.getElementById("year").value;
            let errorMessage = document.getElementById("errorMessage");

            const passkeys = {
                        "1st":
                    "BCA01",
                "2nd": "BCA02",
                "3rd": "BCA03",
                "4th": "BCA04"
            };

            const pages = {
                "1st":
                "bca1.html",
                "2nd": "bca2.html",
                "3rd": "bca3.html",
                "4th": "bca4.html"
            };

            if (passkeys[year] && passkey === passkeys[year]) {
                window.location.href = pages[year]; // Redirects to the corresponding page
            } else {
                errorMessage.textContent = "Incorrect Passkey! Try Again.";
            }
        });
    </script>

</body>
</html>
//All the other departments have different passkeys but the same interface.
The passkeys are as follows:
BBA
"1st": "BBA01",
"2nd": "BBA02",
"3rd": "BBA03",
"4th": "BBA04"
```

```
BCOM
"1st": "BCOM01",
"2nd": "BCOM02",
"3rd": "BCOM03",
"4th":
"BCOM04"
ECONOMICS
"1st": "ECO01",
"2nd": "ECO02",
"3rd": "ECO03",
"4th": "ECO04"
ENGLISH
"1st":
"ENG01",
"2nd": "ENG02",
"3rd": "ENG03",
"4th":
"ENG04" AI
"1st": "AI001",
"2nd": "AI002",
"3rd": "AI003",
"4th": "AI004"//
```

- bca1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Voting Page</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; background-color: #222; color: white; }
      .container { width: 500px; margin: auto; padding: 20px; background-color: #333; border-radius: 10px;
margin-bottom: 30px; }
    .candidate-box { display: flex; justify-content: space-around; align-items: center; margin-top: 20px; }
    .candidate { text-align: center; }
    .candidate img { width: 100px; height: 100px; border-radius: 10px; }
    .vote-options { margin-top: 10px; display: flex; justify-content: center; gap: 20px; }
      .vote-box { display: flex; align-items: center; justify-content: center; width: 120px; height: 40px;
border: 2px solid white; border-radius: 8px; cursor: pointer; transition: transform 0.3s ease-in-out,
background-color 0.3s; }
    .vote-box:hover{transformscale(1.1);background-color:#444;}
```

```
        .vote-box input { margin-right: 5px; }
          .submit-btn { background-color: darkblue; color: white; font-size: 18px; padding: 10px; width: 250px;
border-radius: 8px; font-weight: bold; cursor: pointer; transition: transform 0.3s ease-in-out, background-color
0.3s; }
        .submit-btn:hover { background-color: navy; transform: scale(1.1); }
    </style>
</head>
<body>
    <h2>Voting Page</h2>
    <form action="vote.php" method="POST">
        <input type="hidden" name="page" value="BCA 1 YEAR">

    <!-- General Candidates -->
    <div class="container">
        <h3>General Candidates</h3>
        <div class="candidate-box">
            <div class="candidate">
                <img src="candidate image/bca/bca1/Ayush.jpg"alt="ayush">
                <p>ayush</p>
                <div class="vote-box">
                    <input type="radio" name="generalVote" value="ayush"> Vote
                </div>
            </div>
            <div class="candidate">
                <img src="candidate image/bca/bca1/Bharath.jpg" alt="Bharath">
                <p>Bharath</p>
                <div class="vote-box">
                    <input type="radio" name="generalVote" value="Bharath"> Vote
                </div>
            </div>
        </div>
        <div class="vote-options">
            <div class="vote-box">
                <input type="radio" name="generalVote" value="NOTA"> NOTA
            </div>
        </div>
    </div>

    <!-- Reserved Candidates -->
    <div class="container">
        <h3>Reserved Candidates</h3>
        <div class="candidate-box">
            <div class="candidate">
                <img src="candidate image/bca/bca1/Pooja.jpg" alt="Pooja">
```

```
            <p>Pooja</p>
            <div class="vote-box">
                <input type="radio" name="reservedVote" value="Pooja"> Vote
            </div>
        </div>
        <div class="candidate">
            <img src="candidate image/bca/bca1/Smrithi.jpg" alt="Smrithi">
            <p>Smrithi
            </p>
            <div class="vote-box">
                <input type="radio" name="reservedVote" value="Smrithi"> Vote
            </div>
        </div>
    </div>
    <div class="vote-options">
        <div class="vote-box">
            <input type="radio" name="reservedVote" value="NOTA"> NOTA
        </div>
    </div>
</div>

<button class="submit-btn" onclick="submitVote()">Submit Vote</button>

<script>
    function submitVote() {
        let generalVote = document.querySelector('input[name="generalVote"]:checked');
        let reservedVote = document.querySelector('input[name="reservedVote"]:checked');

        if (!generalVote || !reservedVote) {
            alert("Please select votes for both categories!");
            return;
        }

            alert("Vote Submitted!\nGeneral Candidate: " + generalVote.value + "\nReserved Candidate: " +
reservedVote.value);
    }
</script>
</form>
</body>
</html>
```

//The department interface remains the same, with only the candidate names and details changing. The included pages are **bca2.html, bca3.html, bca4.html, bcom1.html, bcom2.html, bcom3.html, bcom4.html, ai1.html, ai2.html, ai3.html, ai4.html, eco1.html, eco2.html, eco3.html, eco4.html, eng1.html, eng2.html, eng3.html,andeng4.html**.//

❖ **vote.php**

```
<?php

session_start();

error_reporting(E_ALL);

ini_set('display_errors',

1);

// Database connection

$servername = "localhost";

$username = "root";

$password = "";

$database = "user_db";

$conn = new mysqli($servername, $username, $password, $database);

// Check connection

if ($conn->connect_error) {

    die("Connection failed: " . $conn->connect_error);

}

// Ensure user has a unique session-based ID

if (!isset($_SESSION['user_id'])) {

    $_SESSION['user_id'] = uniqid('user_', true); // Generate unique user ID

}

$user_id = $_SESSION['user_id'];

// Check if form data is received

if (!isset($_POST['generalVote']) || !isset($_POST['reservedVote'])) {

    die("Error: Missing form data! Please select both votes.");

}
// Get form data

$page = $_POST['page'];

$generalVote = $_POST['generalVote'];

$reservedVote = $_POST['reservedVote'];
```

```
// Check if user has already voted on this page
$sql_check = "SELECT * FROM votes WHERE user_id=? AND page=?";
$stmt_check = $conn->prepare($sql_check);
$stmt_check->bind_param("ss", $user_id, $page);
$stmt_check->execute();
$result = $stmt_check->get_result();
if ($result->num_rows > 0) {
    echo "<script>alert('You have already voted!'); window.location.href='ai1.html';</script>";
} else {
    // Insert vote into the database
    $sql_insert = "INSERT INTO votes (user_id, page, general_vote, reserved_vote) VALUES (?, ?, ?, ?)";
    $stmt_insert = $conn->prepare($sql_insert);
    $stmt_insert->bind_param("ssss", $user_id, $page, $generalVote, $reservedVote);


    if ($stmt_insert->execute()) {
        echo "<script>alert('Vote submitted successfully!'); window.location.href='thankyou.html';</script>";
    } else {
        echo "Error: " . $stmt_insert->error;
    }
}
// Close connection
$conn->close();
?
```

❖ **results.html**

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Election Results</title>

                <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">

    <style>

        body {

            font-family: Arial, sans-serif;

            background: linear-gradient(135deg, #23232e, #030509);

            color: white;

            text-align:

            center; padding:

            50px;

        }

        .container {

            max-width: 800px;

            margin: auto;

            background: rgba(255, 255, 255, 0.1);

            padding: 60px;

            border-radius: 15px;

            box-shadow: 0px 0px 20px rgba(255, 255, 255, 0.2);

opacity: 0;
```

```
        }
        .btn-custom:hover {
            background-color:
            navy; transform:
            scale(1.1);
        }
        .hidden {
            display:
            none;
        }
        #message {
            font-size: 24px;
            color: #f39c12;
            margin-top:
            20px;
        }
    </style>
</head>
<body>

<div class="container">
    <h2>📢 The results will be published at 12:00 PM.</h2>
    <p>Please check back later.</p>
    <button id="proceedBtn" class="btn-custom">Proceed</button>
    <p id="message" class="hidden">⚠️ Results not published</p>
</div>
<script>
document.getElementById("proceedBtn").addEventListener("click", function() { let now = new Date();
```

```
        let currentHour = now.getHours();
        let currentMinute = now.getMinutes();

        let requiredHour = 12;
        let requiredMinute = 0;

            if (currentHour > requiredHour || (currentHour === requiredHour && currentMinute >=
requiredMinute)) {
            window.location.href = "results.php";
        } else {
            document.getElementById("message").classList.remove("hidden");
        }
    });
</script>
</body>
</html>
```

❖ **results.php**

```
<?php
session_start();
error_reporting(E_ALL);
ini_set('display_errors',
1);
// Database connection
$servername = "localhost";
$username = "root";
$password = "";
```

```
$database = "user_db";

$conn = new mysqli($servername, $username, $password, $database);

// Check connection

if ($conn->connect_error) {

    die("Connection failed: " . $conn->connect_error);

}

// Define department groups

$departments = [

    "BBA" => ["BBA 1 YEAR", "BBA 2 YEAR", "BBA 3 YEAR", "BBA 4 YEAR"],

    "BCA" => ["BCA 1 YEAR", "BCA 2 YEAR", "BCA 3 YEAR", "BCA 4 YEAR"],
    "BCOM" => ["BCOM 1 YEAR", "BCOM 2 YEAR", "BCOM 3 YEAR", "BCOM 4 YEAR"],
      "ECONOMICS" => ["ECONOMICS 1 YEAR", "ECONOMICS 2 YEAR", "ECONOMICS 3
YEAR", "ECONOMICS 4 YEAR"],

     "ENGLISH" => ["ENGLISH 1 YEAR", "ENGLISH 2 YEAR", "ENGLISH 3 YEAR",
"ENGLISH 4 YEAR"],

    "AI" => ["AI 1 YEAR", "AI 2 YEAR", "AI 3 YEAR", "AI 4 YEAR"] // Added AI department

];



// Fetch general category votes

$general_votes_query = "SELECT page AS department, general_vote AS candidate, COUNT(*) AS
total_votes

                FROM votes

                WHERE general_vote IS NOT NULL

                GROUP BY page, general_vote

                ORDER BY page, total_votes DESC";

$general_votes_result = $conn->query($general_votes_query);

// Fetch reserved category votes


$reserved_votes_query = "SELECT page AS department, reserved_vote AS candidate, COUNT(*)
AS total_votes

                FROM votes
```

```
                WHERE reserved_vote IS NOT NULL
                GROUP BY page, reserved_vote
                ORDER BY page, total_votes DESC";
$reserved_votes_result = $conn->query($reserved_votes_query);
// Fetch NOTA votes separately for General and Reserved
$nota_query = "SELECT page AS department,
                SUM(CASE WHEN general_vote = 'NOTA' THEN 1 ELSE 0 END) AS general_nota,
                SUM(CASE WHEN reserved_vote = 'NOTA' THEN 1 ELSE 0 END) AS reserved_nota
            FROM votes
            GROUP BY page";
$nota_result = $conn->query($nota_query);
// Store results
$general_results = [];
$reserved_results = [];
$nota_votes = [];
// Organize general votes
while ($row = $general_votes_result->fetch_assoc()) {
    $general_results[$row['department']][] = $row;
}
// Organize reserved votes
while ($row = $reserved_votes_result->fetch_assoc()) {
    $reserved_results[$row['department']][] = $row;

}
// Store NOTA votes
while ($row = $nota_result->fetch_assoc()) {
    $nota_votes[$row['department']] = $row;
}
```

```
$conn->close();

?>

<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Department-wise Election Results</title>

                    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">

    <style>

        body { font-family: Arial, sans-serif; background-color:rgb(44, 47, 50); padding: 20px; }

        .container { max-width: 1100px; margin: auto; }

        h2, h3 { text-align: center; margin-bottom: 20px; }

        .table { box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.1); }

          .department-section { margin-bottom: 40px; padding: 20px; background: #ffffff; border-radius:
8px; box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.1); }

        .winner { background-color:rgb(91, 134, 93) !important; } /* Green highlight for winners */

    </style>

</head>

<body>

<div class="container">

    <h2 style="color: white !important;">📊 Department-wise Voting Results</h2>

    <?php foreach ($departments as $category => $dept_list) { ?>

        <div class="department-section">

    <h3><?php echo htmlspecialchars($category); ?></h3

            <?php foreach ($dept_list as $dept) { ?>

                <h4><?php echo htmlspecialchars($dept); ?></h4>

                <table class="table table-bordered table-striped">
```

```
<thead class="table-dark">
  <tr>
    <th>Candidate Type</th>
    <th>Candidate 1</th>
    <th>Votes</th>
    <th>Candidate 2</th>
    <th>Votes</th>
    <th>NOTA</th>
  </tr>
</thead>
<tbody>
  <?php
$general_candidates = isset($general_results[$dept]) ? $general_results[$dept] : [];
$reserved_candidates = isset($reserved_results[$dept]) ? $reserved_results[$dept] : [];
$general_candidate1 = isset($general_candidates[0]) ? $general_candidates[0] : ['candidate' => 'N/A', 'total_votes' => 0];
$general_candidate2 = isset($general_candidates[1]) ? $general_candidates[1] : ['candidate' => 'N/A', 'total_votes' => 0];
$reserved_candidate1 = isset($reserved_candidates[0]) ? $reserved_candidates[0] : ['candidate' => 'N/A', 'total_votes' => 0];
$reserved_candidate2 = isset($reserved_candidates[1]) ? $reserved_candidates[1] : ['candidate' => 'N/A', 'total_votes' => 0];
$highest_general_votes = max($general_candidate1['total_votes'], $general_candidate2['total_votes']);
$highest_reserved_votes = max($reserved_candidate1['total_votes'], $reserved_candidate2['total_votes']);
$general_nota = isset($nota_votes[$dept]) ? $nota_votes[$dept]['general_nota'] : 0;
$reserved_nota = isset($nota_votes[$dept]) ? $nota_votes[$dept]['reserved_nota'] : 0;
```

```
                    ?>
  <!-- General Candidates Row -->
                    <tr>
                      <td>General</td>
  <td class="<?php echo ($general_candidate1['total_votes'] == $highest_general_votes) ? 'winner' : ''; ?>">
                <?php echo htmlspecialchars($general_candidate1['candidate']); ?>
                      </td>
  <td class="<?php echo ($general_candidate1['total_votes'] == $highest_general_votes) ? 'winner' : ''; ?>">
                        <?php echo $general_candidate1['total_votes']; ?>
                      </td>
 <td class="<?php echo ($general_candidate2['total_votes'] == $highest_general_votes) ? 'winner' : ''; ?>">
                        <?php echo htmlspecialchars($general_candidate2['candidate']); ?>
                      </td>
  <td class="<?php echo ($general_candidate2['total_votes'] == $highest_general_votes) ? 'winner' : ''; ?>">
                        <?php echo $general_candidate2['total_votes']; ?>
                      </td>
                      <td><?php echo $general_nota; ?></td>
                    </tr>
                    <!-- Reserved Candidates Row -->
                    <tr>
<td>Reserved</td>
<td class="<?php echo ($reserved_candidate1['total_votes'] ==
          $highest_reserved_votes) ? 'winner' : ''; ?>">
<?php echo htmlspecialchars($reserved_candidate1['candidate']); ?>
</td>
  <td class="<?php echo ($reserved_candidate1['total_votes'] == $highest_reserved_votes) ? 'winner' : ''; ?>">
<?php echo $reserved_candidate1['total_votes']; ?>
</td>
```

```
<td class="<?php echo ($reserved_candidate2['total_votes'] == $highest_reserved_votes) ? 'winner' : ''; ?>">
                    <?php echo htmlspecialchars($reserved_candidate2['candidate']); ?>
                  </td>
<td class="<?php echo ($reserved_candidate2['total_votes'] == $highest_reserved_votes) ? 'winner' : ''; ?>">
                    <?php echo $reserved_candidate2['total_votes']; ?>
                  </td>
                  <td><?php echo $reserved_nota; ?></td>
                </tr>
              </tbody>
              </table>
            <?php } ?>
          </div>
        <?php } ?>
      </div>
    </body>
</html>
```

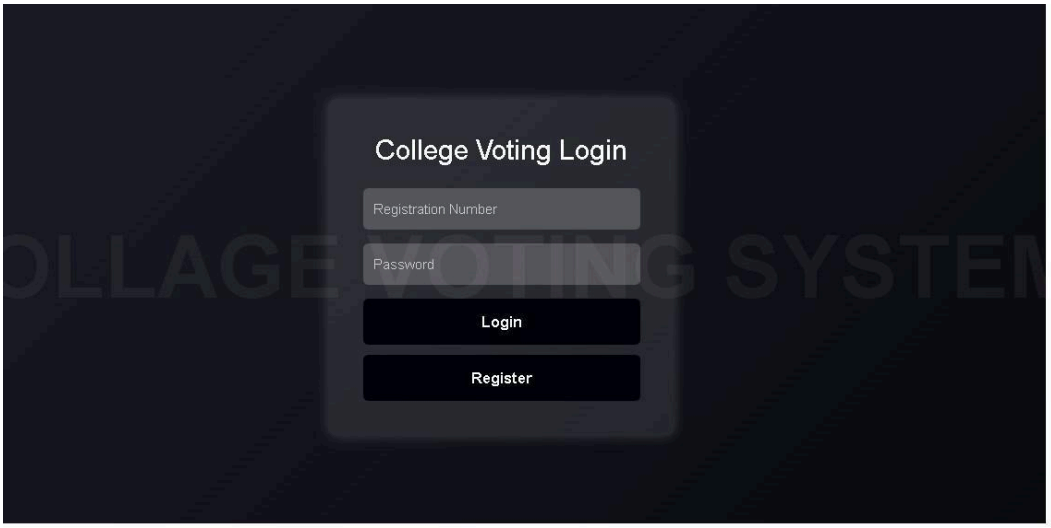
- **logout.php**

```
<?php
session_start();
session_unset();
session_destroy();

if (isset($_SESSION['admin'])) {
   echo "<script>
      alert('Admin successfully logged out!');
      window.location.href = 'adminlogin.html';
   </script>";
} elseif (isset($_SESSION['user'])) {
   echo "<script>
      alert('Student successfully logged out!');
      window.location.href = 'login.html';
   </script>";
} else {
```

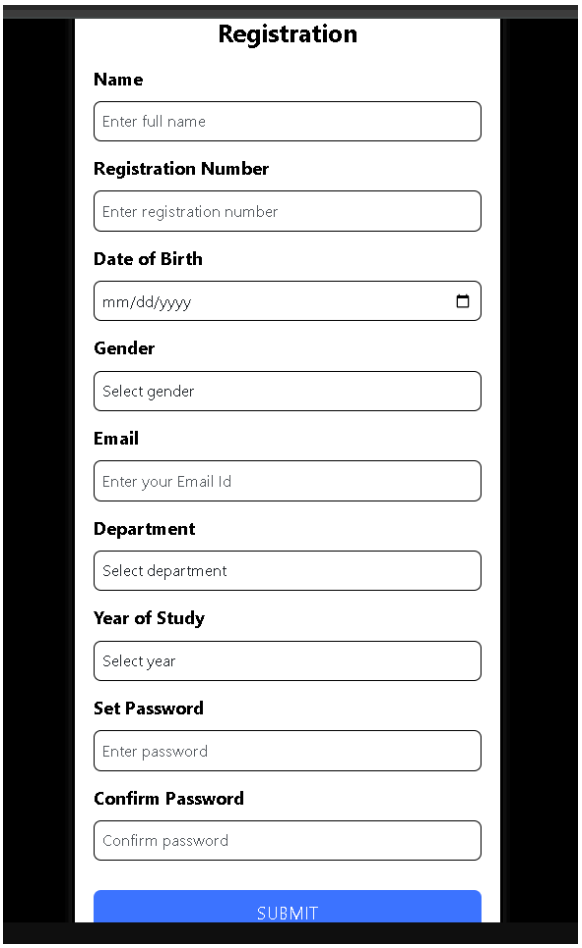
```
  echo "<script>
        alert('Successfully logged out!');
        window.location.href = 'login.html';
    </script>";
}

exit();
?>
```

8.2 GUI SCREEN SHORT

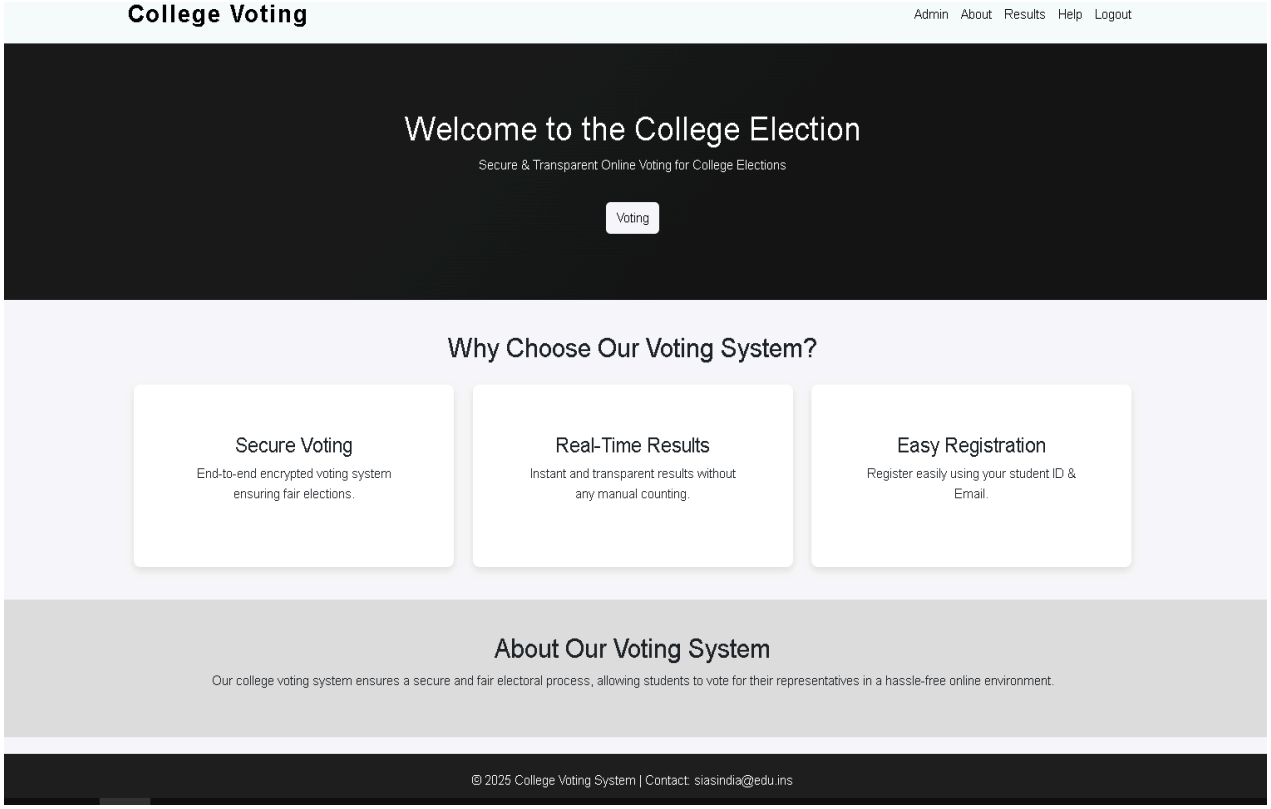
8.2.1 login page



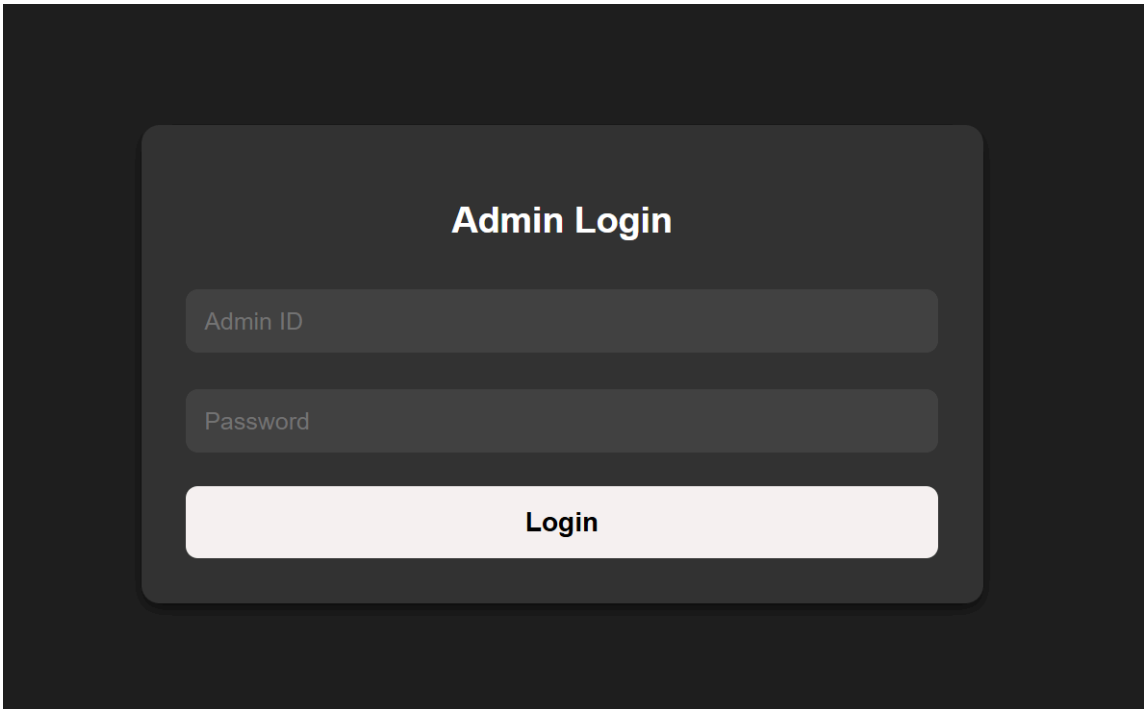
8.2.2 Register page



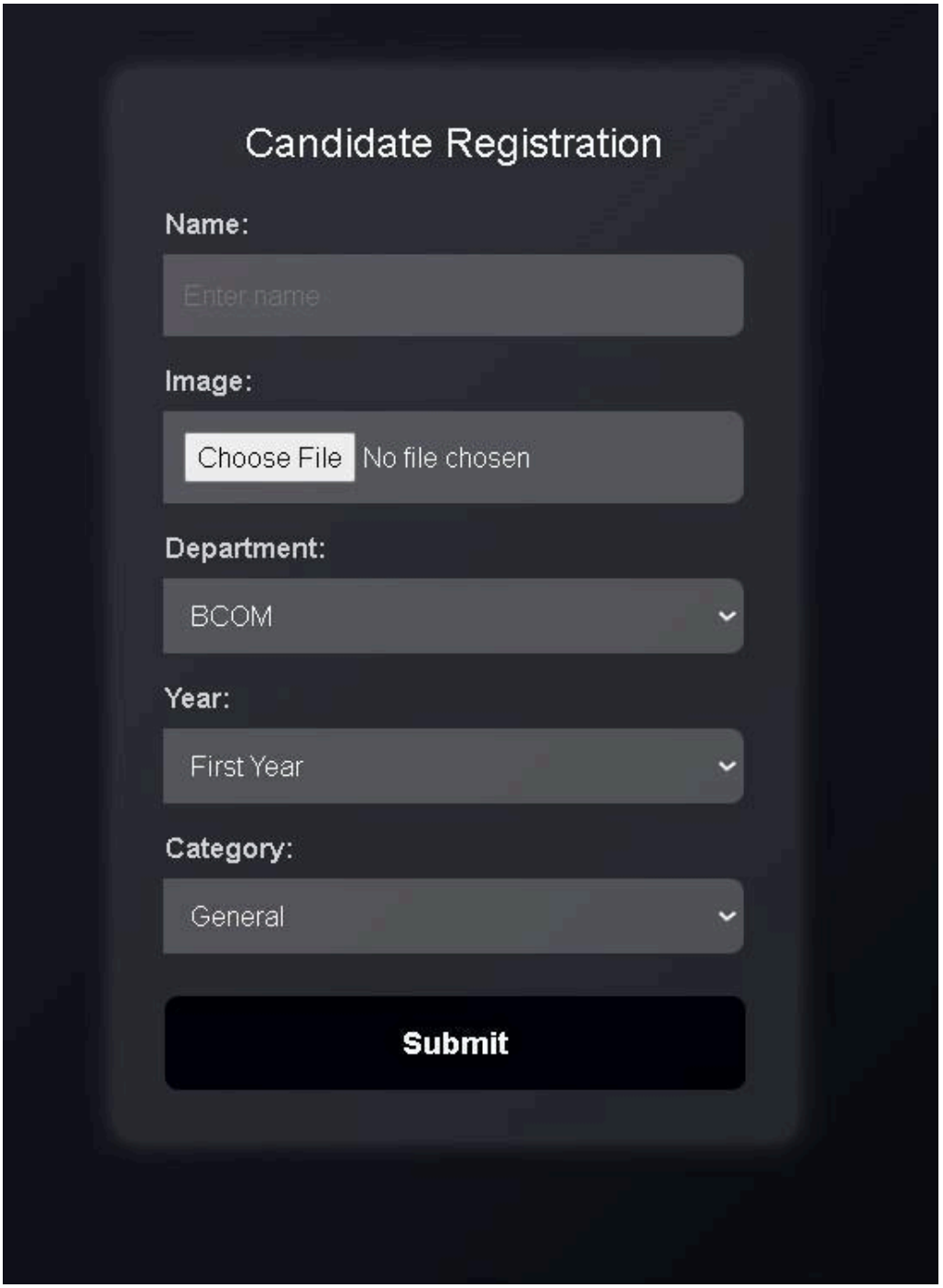
8.2.3 Home page



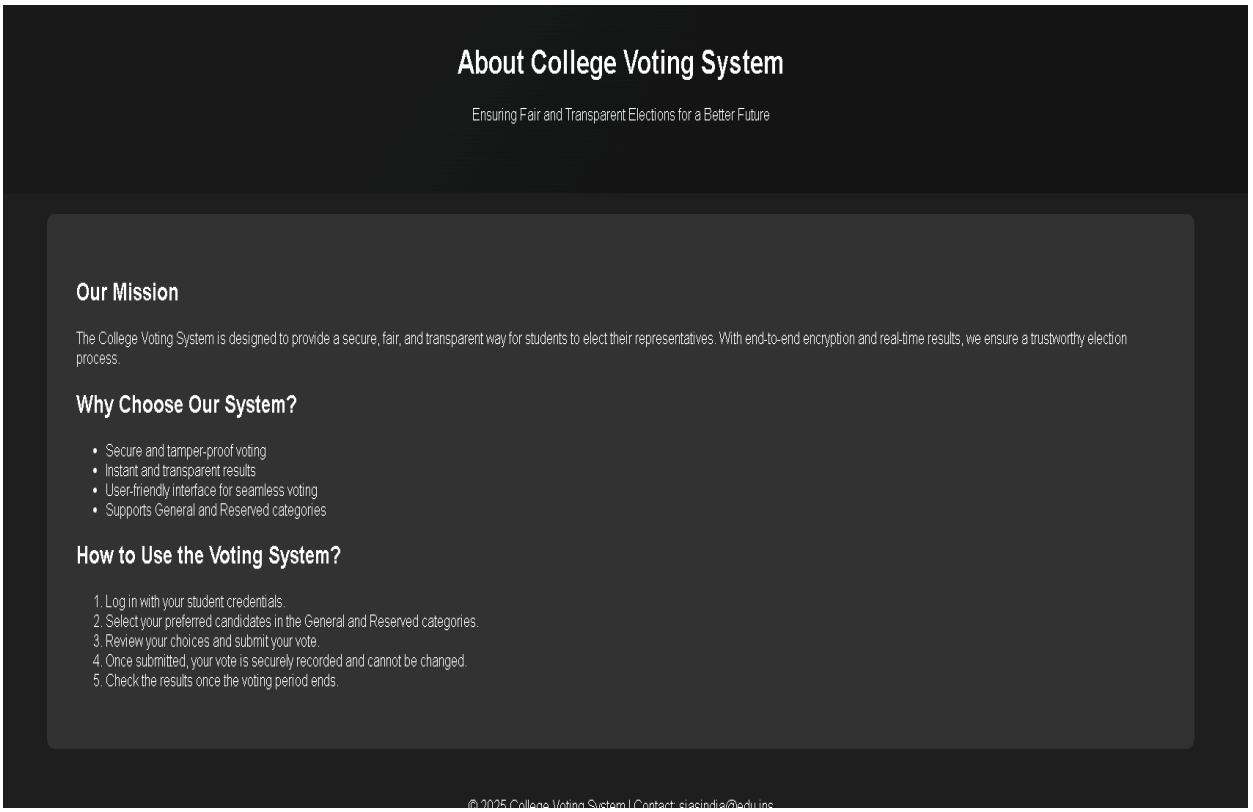
8.2.4 Admin Login



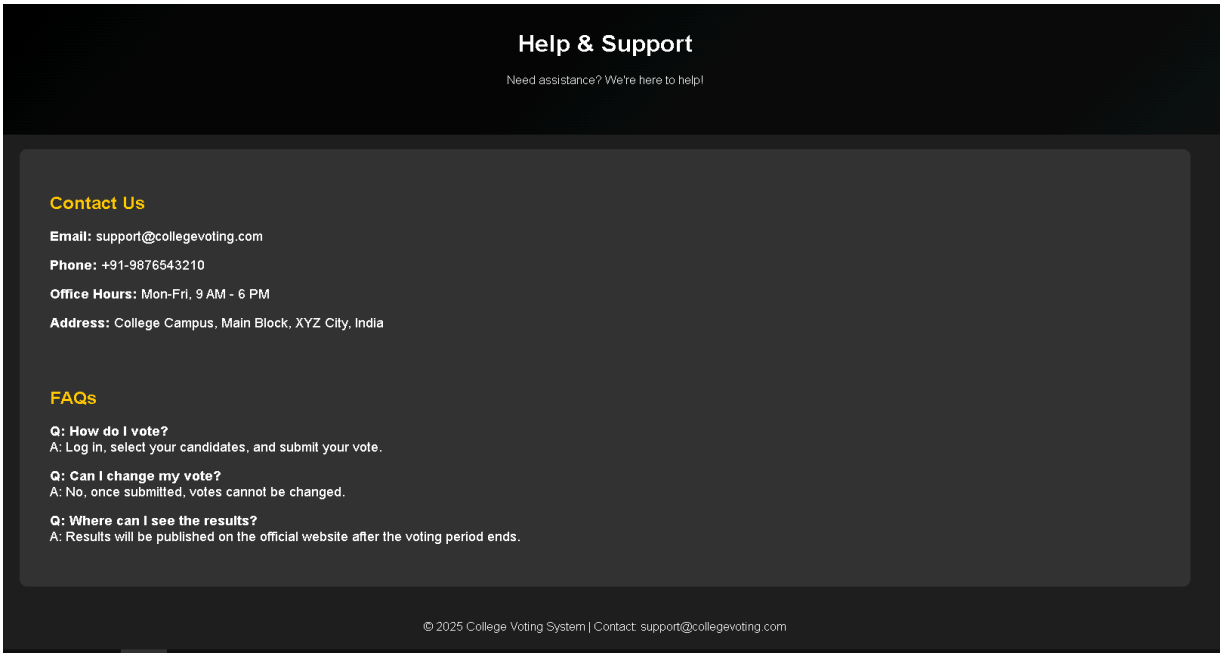
8.2.4 candidate add



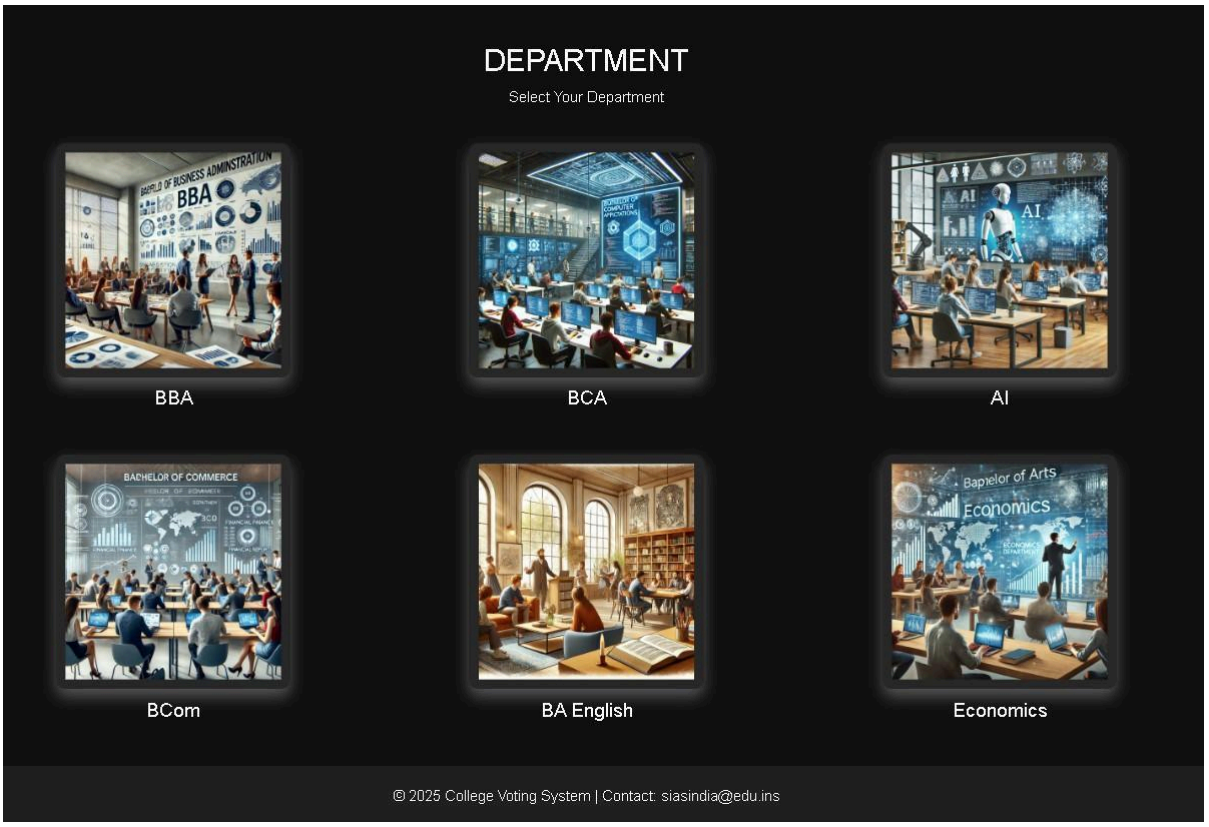
8.2.5 About



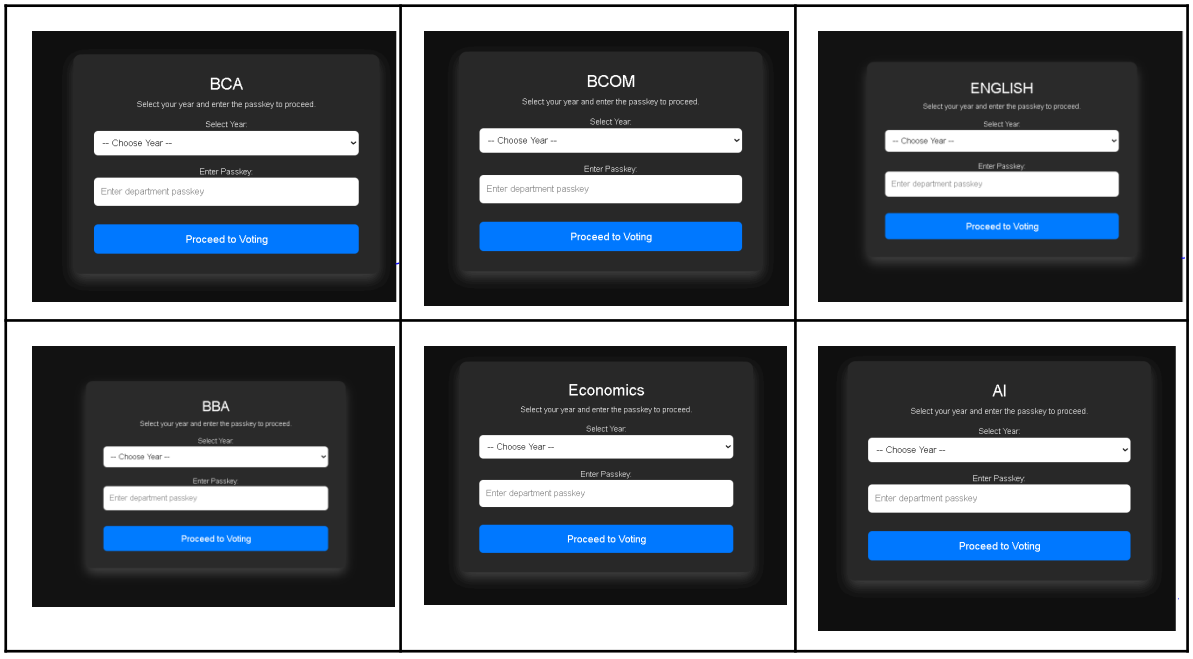
8.2.6 Help & Support



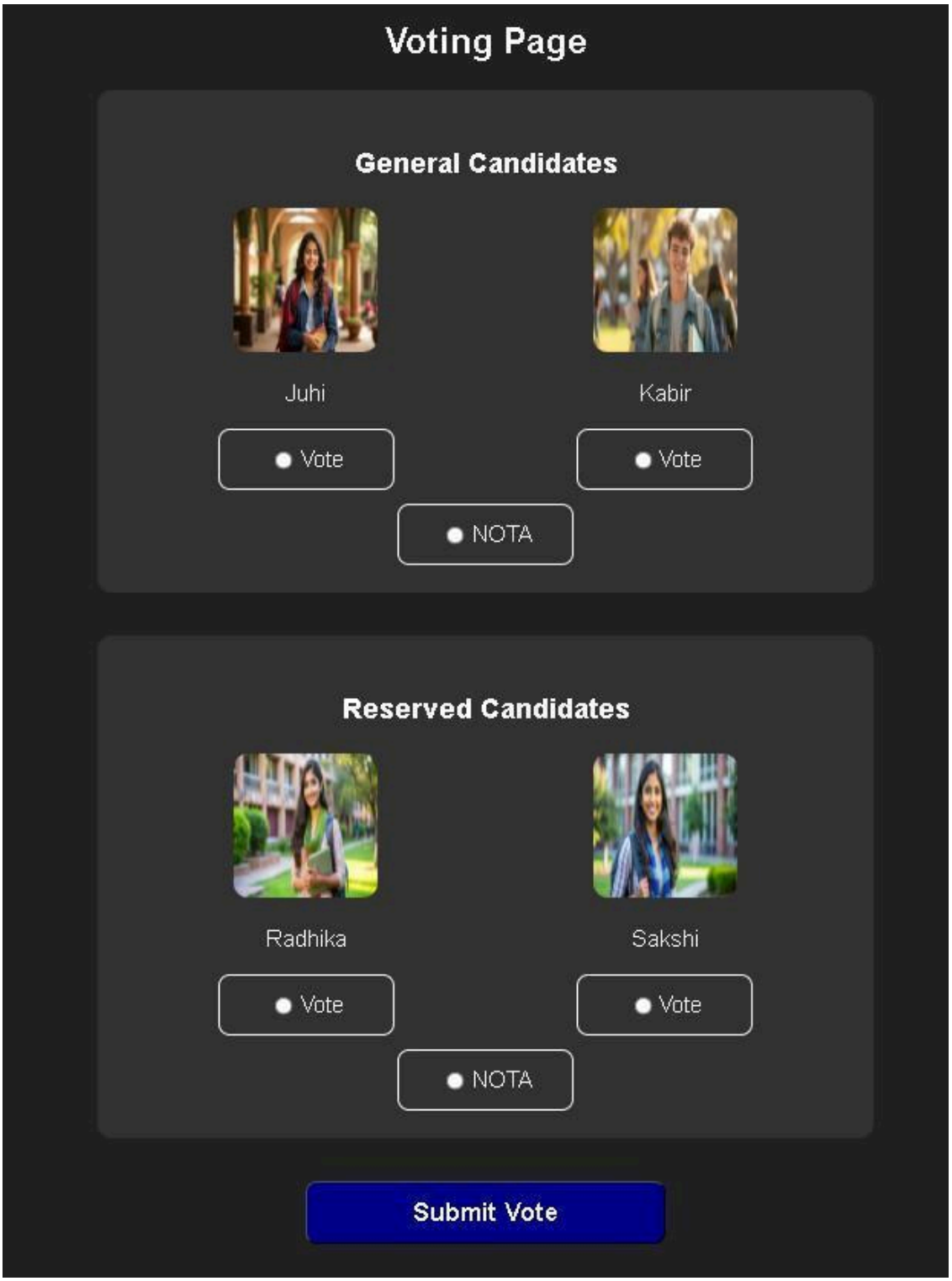
8.2.7 Voting



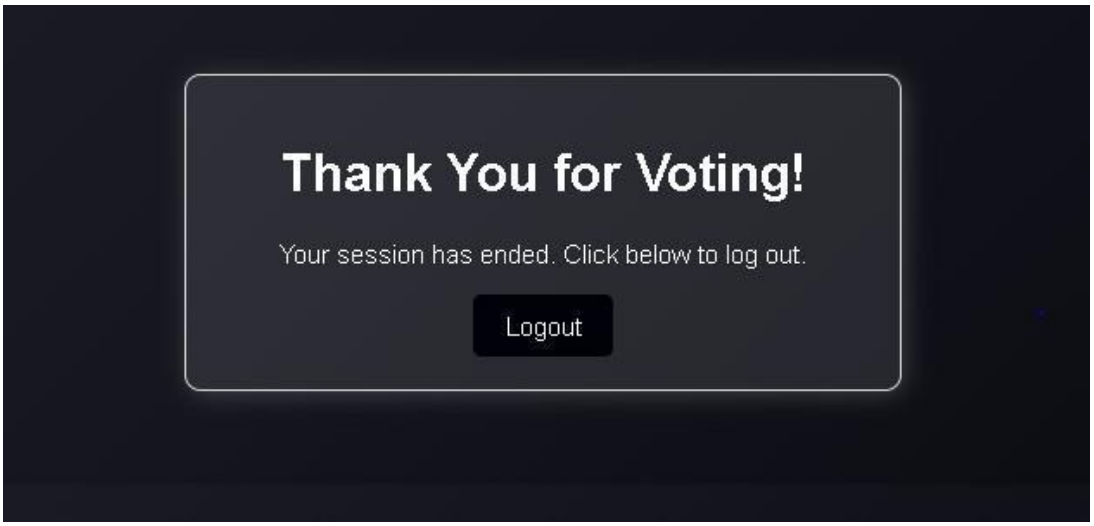
8.2.8 passkey



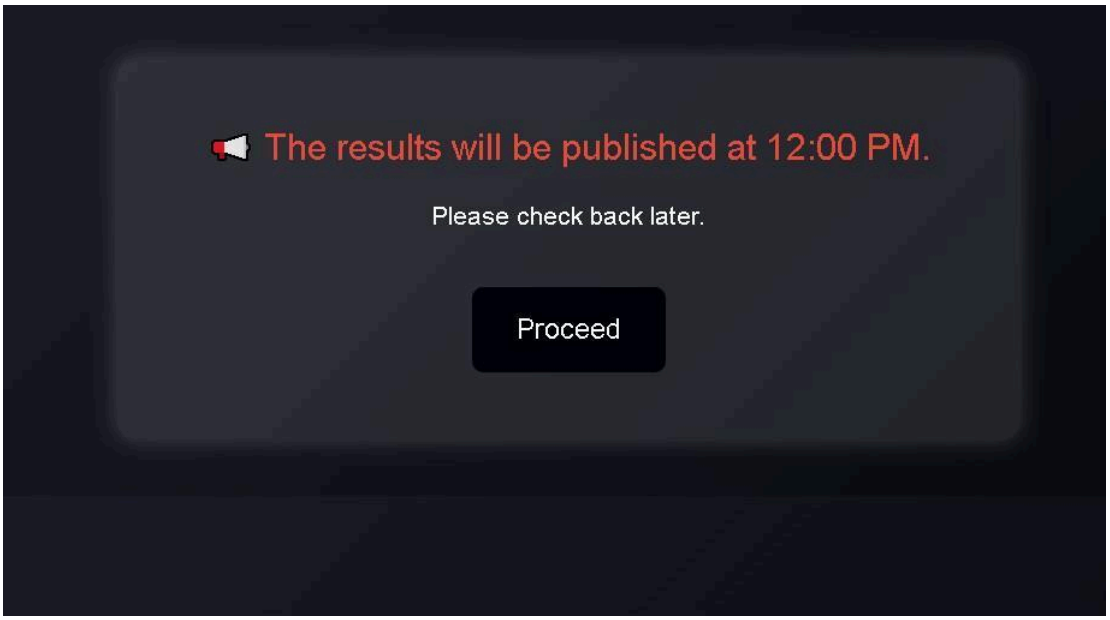
8.2.8 candidate &voting



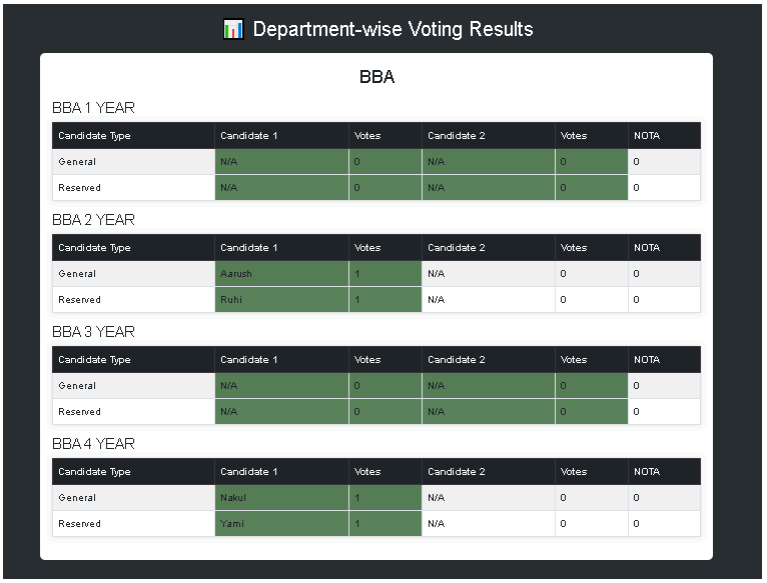
8.2.9 thankyou



8.2.10 result page



8.2.11 result publiished



Department-wise Voting Results

BBA

BBA 1 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	N/A	0	N/A	0	0
Reserved	N/A	0	N/A	0	0

BBA 2 YEAR

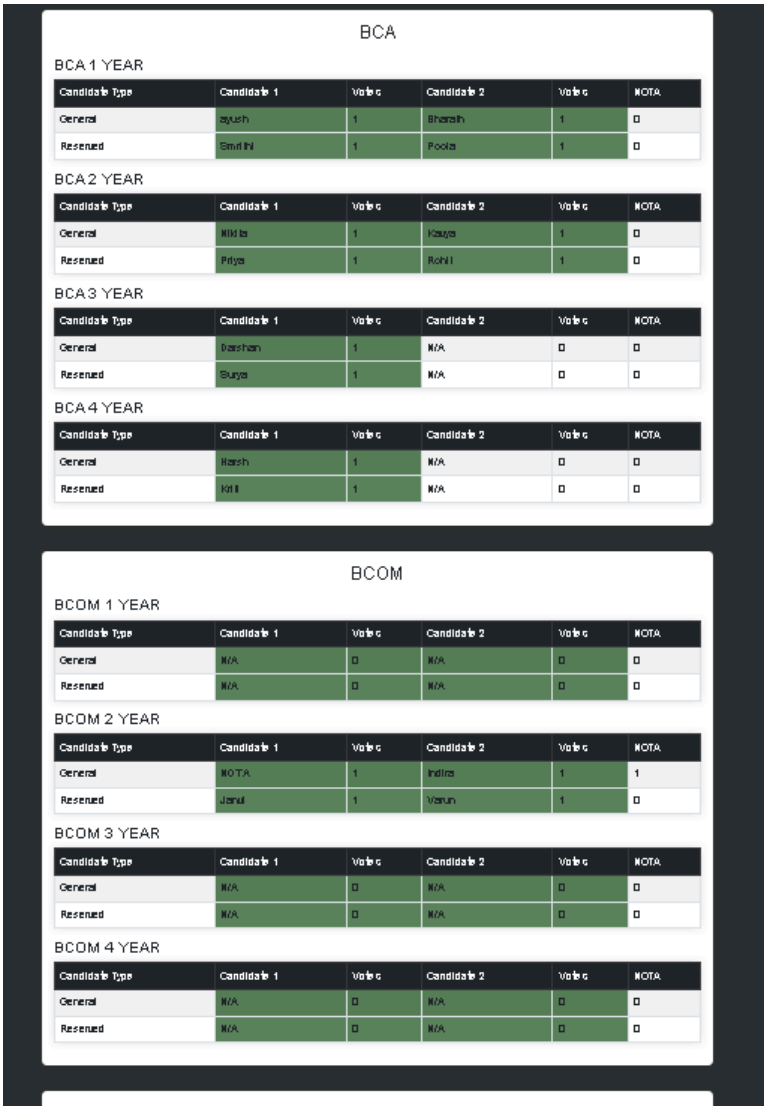
Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	Aarush	1	N/A	0	0
Reserved	Ruhi	1	N/A	0	0

BBA 3 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	N/A	0	N/A	0	0
Reserved	N/A	0	N/A	0	0

BBA 4 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	Nakul	1	N/A	0	0
Reserved	Yami	1	N/A	0	0



BCA

BCA 1 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	Ayush	1	Bharath	1	0
Reserved	Smrithi	1	Pooja	1	0

BCA 2 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	Nikita	1	Kavya	1	0
Reserved	Priya	1	Rohit	1	0

BCA 3 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	Darshan	1	N/A	0	0
Reserved	Surya	1	N/A	0	0

BCA 4 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	Harsh	1	N/A	0	0
Reserved	Kiti	1	N/A	0	0

BCOM

BCOM 1 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	N/A	0	N/A	0	0
Reserved	N/A	0	N/A	0	0

BCOM 2 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	NOTA	1	Indira	1	1
Reserved	Jamal	1	Varun	1	0

BCOM 3 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	N/A	0	N/A	0	0
Reserved	N/A	0	N/A	0	0

BCOM 4 YEAR

Candidate Type	Candidate 1	Votes	Candidate 2	Votes	NOTA
General	N/A	0	N/A	0	0
Reserved	N/A	0	N/A	0	0

8.3 System Minimum Requirements

Software configuration

For the proposed system to work properly, it is necessary that following software are installed and running on the server / client.

Operating system : Windows 7 or above / Linux New

Versions Language : HTML,PHP, JAVA SCRIPT

IDE : Visual Studio code

Front end : HTML, CSS, JAVA SCRIPT AND BOOTSTRAP

Other tools : WAMP, Figma

Back end :PHP,MYSQL

Hardware configuration

It is recommended that for optimal performance, the following minimum hardware are installed.

Processor : Intel Pentium IV or above

Monitor : Minimum 14-inch

display RAM : 4 Gb

Hard Disk : 80 GB

Keyboard : Standard 104

Keys Mouse : Serial mouse

9.REFERENCES

9.1 Book References:

1 Robin Nixon – Learning PHP, MySQL & JavaScript

2 Jon Duckett – HTML & CSS: Design and Build Websites

3 Val Head – Designing Interface Animation

9.2 Website References:

1. W3Schools – HTML, CSS, JavaScript Basics https://www.w3schools.com

2. GeeksforGeeks – PHP, MySQL, and Web Development https://www.geeksforgeeks.org

3. GitHub – Sample Voting System Projects https://github.com/search?q=voting+system

4. OWASP – Security Guidelines for Online Voting https://owasp.org/www-project-top-ten/